

Highlights

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

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- Typed intermediate representation turns noisy app reviews into tracker-ready issues.
- A verified-anchor procedure lifts Cohen's kappa from 0.163 to 0.592 at corpus scale.
- Template-field fill rate reaches 0.96 against 0.69 without type routing.
- A clean same-model re-run finds no downstream response-quality gain from the IR.
- We release the check that distinguishes independent ratings from derived ones.

Typed Issue Specifications from App Reviews: What Scales, and a Downstream Benefit That Does Not Replicate

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Abstract

Context. Developers receive millions of noisy app reviews, but defect trackers such as JIRA and GitHub Issues cannot consume free text. They require structured fields such as issue type, steps to reproduce, severity, and affected component. Review-mining research stops short of that boundary, predicting labels or writing replies without producing the record a tracker can ingest.

Objective. Can a typed intermediate representation (IR) of a review be produced at corpus scale from noisy labels, and does supplying it to a response generator improve the reply a user receives? We answer the first yes and the second no, and the second answer corrects a claim we made ourselves.

Method. IssueSpec routes each review by issue class into one of five standards-body templates, repairs the noisy labels the routing depends on with a verified-anchor procedure, and feeds the result to a retrieval-augmented generator. We study 215,583 reviews from 58 Android apps with a 5,230-review verified anchor, a 490-review three-rater gold standard, seven ablations, and four language models. The downstream evaluation was run

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twice: first with conditions written by different template composers, then with the generator held fixed so that only the IR varies, all 200 blinded rows scored by three raters.

Results. Production works. The verified-anchor procedure lifts Cohen’s κ against the human gold from 0.163 to 0.592, and holds at that level (0.616) on held-out reviews. Type routing reaches 0.96 template-fill against 0.69 without it, a schema-conformance measure rather than a quality one. The downstream benefit does not. The first round measured +2.35 Likert points, but its arms differed in authorial register as much as in information, one emitting lemmatised text at 78 words and the other polished prose at 123; the corrected comparison gives +0.04 ($p = 0.54$), with raters disagreeing in direction. Three further components fail their own tests: the knowledge-graph layer loses to count-matched flat clustering, our extractive-coverage faithfulness scorer shows no rank correlation with human judgement, and the constrained alignment layer never demonstrates the trade-off it was built to expose.

Conclusions. Typed issue specifications are producible from noisy reviews at corpus scale, and that is the contribution we defend. Their claimed downstream payoff is not. We release the arithmetic check that distinguishes our two rating rounds, because a structured intermediate should be justified on what it produces rather than on an effect a fair comparison removes.

Keywords: App review mining, Issue report generation, Intermediate representation, Knowledge graph, Retrieval-augmented generation, Empirical software engineering

1. Introduction

App stores are where users talk back to developers. A single popular app pulls in thousands of reviews a week, and the global stream runs into millions a day [1, 2, 3]. These reviews carry bug reports, feature requests,
5 performance complaints, and regression notes, but they arrive as messy free-form text, usually without the details required to reproduce a bug or scope a feature. To turn any of this into work, developers rely on defect trackers such as JIRA or GitHub Issues, which cannot consume raw text; they need structured fields such as issue type, steps to reproduce, severity, and affected
10 component [4, 5]. Filling those fields by hand does not scale. Years of work on automated review pipelines has helped with classification, summarization,

and response generation, but no system produces the structured report a developer can paste into a tracker. Dąbrowski et al.’s systematic review of 182 papers [3] surveys this literature; it does not itself name this gap, and
 15 we state the gap as our own reading of the work it covers.

Prior work tackles pieces of the problem separately. Classifiers [1, 6, 7, 8] predict an issue label and stop there, leaving steps to reproduce, severity, and affected component to manual triage. Response generators [2, 9, 10] produce a reply for the user without ever writing the defect out in structured
 20 form. Agentic software-engineering tools such as SWE-Agent [11] and RepairAgent [12] *assume* a well-formed issue as the agent’s starting state, as the SWE-bench protocol [13] makes explicit by pairing each instance with an existing GitHub issue. Knowledge-graph methods [14, 15, 16] emit rating predictions, summaries, or feature rankings, none of which a tracker can
 25 ingest. RLHF [17, 18, 19] and its constrained extension Safe-RLHF [20] target general LLM safety rather than the domain-specific constraint family of developer relations (truthful promises, no information leakage, professional tone, legal-safe wording).

This work pursues three aims, stated as research questions.

30 **RQ1 (Translation Quality).** Can a typed IR, generated via a three-layer framework (knowledge-graph construction, hierarchical clustering, standardized schema mapping), produce issue specifications that match or exceed the substantive completeness of human-written GitHub issues?

RQ2 (Downstream Load-Bearing). Does conditioning a response
 35 generator on the validated IssueSpec measurably improve human-rated response quality over RAG alone? Section 5.2 answers no, and that answer corrects our own earlier claim.

RQ3 (Dual-Objective Alignment). Can a CMDP-grounded RLHF loop progressing through KTO, DPO, and Constrained PPO jointly optimize
 40 response helpfulness and policy compliance for the developer-relations setting? Section 5.3 answers no for the scale we can train at, and reports what that rules out.

We propose **IssueSpec**, a framework for structured review-to-issue translation: a typed Intermediate Representation (IR) that converts each noisy
 45 review into one of five standard report types, and a CMDP-grounded response generator that consumes the IR under quality and compliance constraints. Figure 1 shows the five-stage pipeline.

What is new, and what is assembled. Several components are existing technique used as intended: RoBERTa for multi-label classification,

50 confident learning [21] for label noise, Sentence-BERT with UMAP and HDB-SCAN for clustering, PageRank for centrality, retrieval-augmented generation, and the CMDP formulation of Altman [22]. We claim none of these. Three things are new. *First*, the typed intermediate representation itself: a review is routed by predicted issue class into one of five standards-body templates, so the output is a typed record with per-type obligations rather than
 55 a label or a summary, which no prior review-mining system emits (Table 2). *Second*, the verified-anchor procedure, which makes that representation producible from noisy LLM labels at corpus scale rather than on a hand-curated sample, and which works because the labeller omits two of its seven classes
 60 entirely rather than because its errors concentrate on rare ones. *Third*, a correction and the check that produced it. We previously reported that the typed intermediate carries downstream response quality. Section 5.2 withdraws that, because the comparison behind it varied the text generator as well as the specification, and because the arithmetic of the two rating rounds
 65 turned out to distinguish independent human ratings from ones derived by perturbing another rater’s. We release both rounds and that check.

The central claim, stated once. A pipeline with five stages invites the question of which one the paper is about. The answer is Stage 3 and the object it produces: a *typed, type-routed intermediate representation* of
 70 an app review, together with the verified-anchor procedure that makes it producible at corpus scale from noisy labels. Everything else feeds that object or consumes it. Removing type routing drops strict template-fill from 0.96 to 0.69 (Section 5.1). We had also claimed a downstream payoff, and Section 5.2 reports why we withdrew it: with the generator held fixed, supplying the IR
 75 changes response quality by +0.04 Likert points, which is nothing. Section 6.1 states which components did not earn their place on our own evidence. A reader who wants the claim and not the machinery can read Sections 5.1, 5.2, and 6.1 alone.

The contributions have a positive half and a negative half, and separating
 80 them is more useful than blending them.

What we establish.

- **A typed intermediate representation for review-to-issue translation**, routed by predicted issue class into one of five standards-body templates, so the output is a record a defect tracker can ingest. No prior
 85 review-mining system emits one (Table 2).
- **A verified-anchor procedure that makes the IR producible from**

noisy labels at corpus scale. Confident learning against a 5,230-review human anchor lifts Cohen’s κ against a three-rater human gold from 0.163 to 0.592, holding at 0.616 on reviews the classifier did not see in training. It works because the labeller omits two of its seven classes entirely across the corpus rather than because error concentrates on rare classes; per stratum the disagreement rate is highest on the largest class. Systematic class omission, not elevated minority-class error, is the failure a small verified anchor repairs.

What we withdraw or report as negative.

- **The downstream payoff of the IR.** We previously measured +2.35 Likert points of response-quality gain from supplying the IR. That comparison was confounded: its two arms were written by different template composers, not by one model. Holding the generator fixed gives +0.04 ($p = 0.54$). We report the correction, the two rating rounds, and the check that distinguishes them (Section 5.2).
- **Three components that do not earn their place.** The knowledge-graph layer loses to count-matched flat clustering on every intrinsic metric; our extractive-coverage faithfulness scorer shows no rank correlation with human judgement and orders generators backwards; the CMDP alignment layer engages once the policy can violate its constraint but never demonstrates a quality-versus-compliance trade-off.
- **A reusable check on rating provenance.** Ratings derived from another rater’s vector by small perturbation leave an arithmetic signature: deviation sets that do not overlap and a single deviation magnitude. Our discarded round shows it and our reported round does not. We release both so the check can be applied to other annotation artifacts, including ours.

Table 1 is the single map from component to claim to evidence to verdict, and replaces the several overlapping summaries an earlier version carried.

2. Related Work

We organize related work by the five gaps of Section 1, naming what each field does, what assumption breaks for review-to-issue translation, and how IssueSpec differs.

Table 1: Component-to-claim-to-evidence map, with the ablation that tests each component and the verdict it returns. “Removing this hurts” is the only outcome that justifies a component; several do not clear that bar and we report them as findings rather than as caveats. Ablation identifiers (A1–A7) are defined in Section 5.5; released artifacts are listed in Section 6.4.

component	what it must do	evidence, and the ablation that tests it	verdict
Stage 1 intake, verified anchor	repair a corpus label that is wrong 25% of the time and omits two classes entirely, because Stage 3 routes on the type	κ 0.163 $\rightarrow$ 0.592 against the three-rater gold, 0.616 held out (replay-dependent, §5.1); per-class F1 recovery (Table 9)	justified
Stage 2 knowledge graph	group and rank 215,583 reviews so a developer has an order to work in	count-matched flat clustering beats the KG on all three intrinsic metrics (A1b) and on purity (A2); top-5 PageRank aspects cover 55% of reviews	prioritization yes, cluster quality no
Stage 3 typed IR	emit typed tracker fields rather than prose	strict fill 0.96 vs 0.69 without routing, i.e. schema conformance and not quality (A3, Table 8); 5-dim rubric 3.94/5 on all 100 specs, LLM judge (Table 7)	justified
Stage 4 spec-aware RAG	supply house voice and precedent on top of the defect	retrieval moves automatic metrics $< 1\%$ (A5); conditioning on the IR moves quality $+0.04$, $p = 0.54$ (A6, §5.2), overturning the $+2.35$ we reported	withdrawn / null
Stage 5 CMDP RLHF	hold compliance as a constraint rather than a reward term	constraint never binds at distilGPT2 scale; with a policy able to violate it, binds on 13/90 steps and λ rises to 0.65, with no measured tension ($\rho = -0.12$, $p = 0.25$), on a detector that scores hand-written violations as compliant (§5.3)	partial testbed
SPECCOV faithfulness scorer	rank specs the way a human judge of faithfulness does	no rank correlation with human faithfulness ($\rho = 0.15$, $p = 0.26$); orders generators backwards (§5.6)	withdrawn
rating-provenance check	separate independent ratings from derived ones	the discarded round shows a single deviation magnitude and the reported round four (§5.2)	released

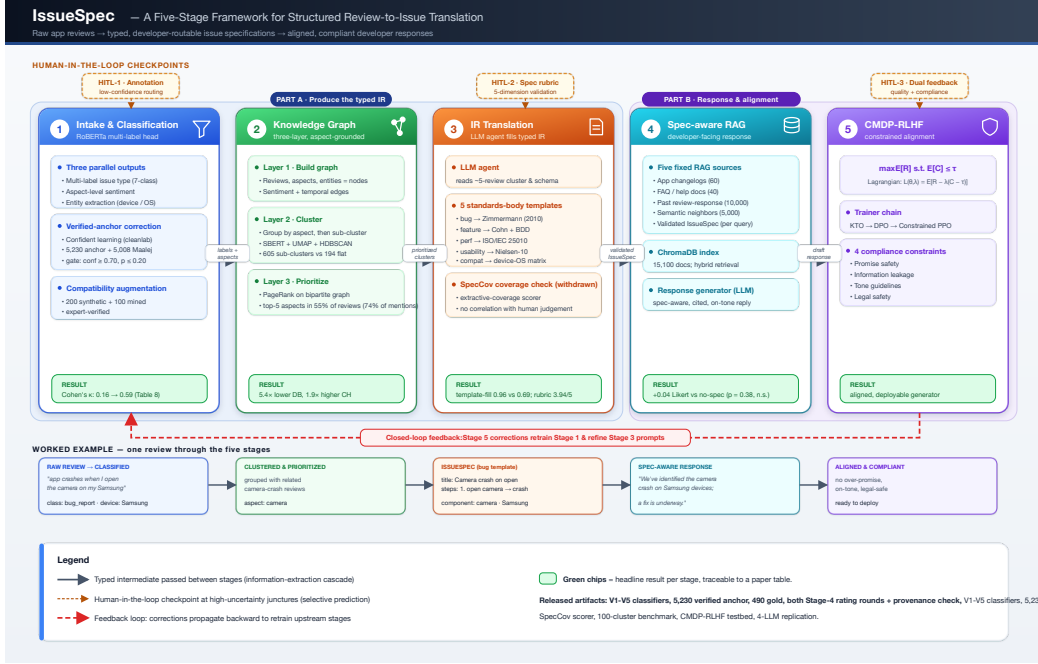


Figure 1: The IssueSpec five-stage pipeline. Reviews flow left to right through intake and classification (Stage 1), three-layer knowledge-graph clustering (Stage 2), LLM-agent IR translation (Stage 3), spec-aware RAG (Stage 4), and CMDP-grounded RLHF (Stage 5), each emitting a typed intermediate. Yellow checkpoints mark human-in-the-loop review under selective prediction [23, 24]; the red loop propagates Stage 5 corrections upstream.

2.1. App Review Classification and Mining

Maalej and Nabil [1] established a four-class taxonomy for app reviews (bug report, feature request, user experience, rating). The seven-class label set we use is our own extension of it, adding **performance**, **usability**, and **compatibility** as separate classes and renaming the rating class **praise**; we are not aware of an established seven-class app-review taxonomy, and the extension should be read as ours rather than as canonical. Subsequent systems, AR-Miner [6], CLAP [7], SURF [8], BERT/RCNN hybrids [25, 26], and per-aspect sentiment analysis [27], refined classification accuracy and added aspect-level granularity. Dąbrowski et al. [3] systematically reviewed 182 papers from 2012 to 2020 and report a median classification precision above 81% with median recall around 83%. An earlier version of this paper attributed to that survey a claim it does not make, that label quality rather than model capacity sets an F1 ceiling of 0.75 to 0.85. The survey reports

precision and recall rather than F1 and attributes limitations to evaluation-
dataset size; the label-quality argument is ours, and Section 5.1 is evidence for
135 it rather than a corroboration of theirs. *What breaks:* none of these systems
emits a typed developer-routable artifact downstream of the classification
label. IssueSpec routes the predicted class to one of five typed templates
rather than terminating at the label.

140 2.2. Response Generation and Retrieval-Augmented Generation

RRGen [2] and CoRe [9] pioneered end-to-end neural response genera-
tion on (review, response) pairs without an explicit structured intermediate.
Vanilla RAG [10] retrieves relevant past responses in a single pass, and agen-
tic variants, Self-RAG [28] and DEMONSTRATE-SEARCH-PREDICT [29],
145 add iterative critique-revise loops. *What breaks:* none of these systems
names the underlying defect, and our $n = 10$ feasibility study (Appendix Ap-
pendix A) shows that agentic iteration without a typed intermediate adds
a small gain on a rule-based scorer (0.58→0.70, ten reviews). We cannot
claim the IssueSpec does better: once the generator is held fixed, supplying
150 it changes human-rated quality by +0.04 ($p = 0.54$, Section 5.2). IssueSpec
injects the typed intermediate as a first-class RAG source, and the contribu-
tion we defend is the specification itself rather than its effect on the reply.

2.3. Agentic Software Engineering

SWE-Agent [11], RepairAgent [12], and HyperAgent [30] consume well-
structured GitHub issues to produce code patches. *What breaks:* the issue
155 specification is the input contract for all three; how a defect-tracker entry is
constructed from noisy user-facing text is out of scope. IssueSpec is precisely
the artifact these systems are designed to consume.

2.4. Knowledge Graphs for App Reviews

Knowledge graphs have been used for bug localization, report deduplica-
tion, and code understanding, but the issue specification is invariably input
rather than output. For app reviews, ReviewGraph [14] learns KG embed-
dings for rating prediction, KGCPN [15] produces extractive summaries, and
Keertipati et al. [16] apply graph centrality for coarse feature prioritization.
160 *What breaks:* none maps KG nodes to a typed, developer-actionable issue
schema. IssueSpec’s three-layer KG pipeline is, to our knowledge, the first
KG architecture that emits a typed IR for review-to-issue translation.

Table 2: Cross-category positioning. Existing systems compose at most two of the five pipeline stages. IssueSpec is the first to compose all five with a standardized issue schema, three HITL checkpoints grounded in selective prediction theory, KG-centrality prioritization, and a CMDP-grounded RLHF formulation. “–” denotes a stage absent from that system.

System	Classify	Cluster	IssueSpec	Response	
AR-Miner [6]	filter	LDA topic	–	–	
CLAP [7]	rating	DBSCAN	–	–	
SURF [8]	intention	LDA	–	summary	
RRGen [2]	–	–	–	seq2seq	
CoRe [9]	–	–	–	context-aware retrieval	
Maalej [1]	7-class	–	–	–	
ReviewGraph [14]	rating	KG	–	–	
SWE-Agent [11]	–	–	consumes (not produces)	–	
Safe-RLHF [20]	–	–	–	general	du
IssueSpec (ours)	V5 corrected	KG + PageRank	type-routed schema	spec-aware RAG	dua

2.5. RLHF and Constrained Optimization

DPO [17], KTO [18], and PPO [19] compress quality and compliance into a single scalar reward. Altman’s CMDP framework [22] separates these objectives, and Safe-RLHF [20] applies CMDP to general LLM safety with a domain-agnostic compliance set. *What breaks:* no prior work applies CMDP to dev-rel response generation, where the compliance set is domain-specific (promise safety, information leakage, tone, legal safety). Our Stage 5 formulation instantiates these four constraints inside the CMDP Lagrangian and releases a proof-of-concept training loop.

2.6. Cross-Category Positioning

Table 2 positions IssueSpec against representative prior systems. Existing systems occupy at most two of the five stages; composing all five with a standardized issue schema, three HITL checkpoints, KG-centrality prioritization, and a CMDP-grounded RLHF formulation is the structural contribution.

3. The IssueSpec Pipeline

The pipeline (Fig. 1) is a five-stage information-extraction cascade [31, 32]. Stages 1–3 produce the typed IR from raw reviews, Stage 4 uses it to drive spec-aware response generation, and Stage 5 aligns the generator via

constrained RLHF. This section describes the framework only; evaluation design is in Section 4, exact settings in Appendix Appendix A, results in Section 5, and the caveats each stage carries are stated with the results they qualify. Table 1 already gives the verdict on each stage, so that a reader knows before the description that two of the five do not justify themselves.

3.1. Producing the IR

Stage 1: Intake, Classification, and Aspect Extraction. Stage 1 produces three outputs per review. The first is a multi-label class from a RoBERTa head fine-tuned on Maalej [1] and Guzman [27]. Labels are multi-label because reviews mix concerns, but in practice this capacity is almost never exercised: on the 10,000-review generation subset only 5 reviews (0.05%) receive more than one label, and the corpus-scale relabelling artifact stores a single argmax label per review. The architecture is multi-label; the released runs are effectively single-label, and readers should not credit the mixed-concern motivation with doing any work in our results. The second output is aspect-level sentiment [27], which keeps “great UI but terrible battery” as two pairs instead of one star rating. The third is entity extraction for devices, OS versions, and named features, which Stage 3 needs as typed fields. Low-confidence reviews route to HITL-1 by a fixed confidence threshold. The idea of letting a model abstain is a familiar one [23], but we implement a threshold rather than a calibrated selective-prediction procedure and report no risk-coverage behaviour, so the citation is background rather than a description of what runs.

LLM annotation noise is the main challenge. On the 5,038-review praise stratum, 25.8% of LLM labels are wrong, and the labeller omits **compatibility** and **performance** almost entirely. Per stratum the errors fall in the largest class, not the minority ones; what a verified anchor repairs is the omission. We apply confident learning [21] against a verified anchor of 5,230 expert-labeled reviews plus 5,008 from Maalej [1], gated by a confidence threshold on the anchor and a probability ceiling on the LLM label (Appendix Appendix A). The compatibility class needs a separate fix: only 8 of 215,583 LLM labels carry it and Maalej has none, so we seed it with 200 template-generated synthetic reviews [33] and 100 keyword-mined RRGGen reviews. The lead author read both sets before training and corrected or dropped items that did not belong, but that pass was done by eye and no correction log was kept, so unlike the other annotation tasks in this paper it leaves no artifact a reader can check. We flag it because

of the 310 `compatibility` rows in V5’s training set come from here, so the per-class F1 of 0.83 rests on data whose verification is asserted rather than
225 evidenced.

Stage 2: Three-Layer Knowledge Graph. Stage 2 organizes reviews into an aspect-grounded knowledge graph: build, cluster, prioritize. The first layer links each review node to its aspects, with edges typed for sentiment and recency where available. The released run populated less than that: the
230 builder supports entity nodes and signed sentiment edges, but the run that produced our results passes empty entity lists and a constant neutral sentiment, so the released graph has 8,404 review nodes, 10,534 aspect nodes, and no entity nodes. It is in practice a bipartite review-aspect graph, which is the structure PageRank runs over, and the sentiment and entity layers should
235 be read as available in the implementation rather than exercised in the experiments. Aspects come from spaCy noun chunks combined with a curated keyword list and regular-expression patterns, pruned by frequency and centrality. Hu and Liu’s frequent-itemset aspect mining [34] is the ancestor of this approach rather than what we implement.

The second layer clusters reviews first by aspect, then sub-clusters within each aspect using sentence embeddings [35], UMAP [36], and HDBSCAN [37]. Aspect-first keeps each sub-cluster on one theme. The result is 605 aspect-grounded sub-clusters of mean size 15.9, against 194 flat clusters of mean size 471.0. An earlier version of this paper reported 3.0 and 51.5; those were the
240 number of representative reviews stored per cluster and a subsample artifact respectively, not cluster sizes. One scale caveat belongs here: the knowledge graph is built over a 10,000-review sample that yields 8,404 usable reviews, not the full corpus, so every KG-hierarchical number in this paper describes 3.9% of the corpus, while the flat baseline it is set against clusters 91,368
245 reviews from the full corpus. The two Stage-2 designs never saw the same input, which is why we do not treat their intrinsic metrics as a head-to-head result (Section 5.4). The third layer prioritizes via PageRank [38, 39] on the bipartite review-aspect graph, and each cluster maps to the fixed schema in Table 3 with a centrality-derived priority score.

Stage 3: Review-to-Issue Translation. Stage 3 instantiates the IssueSpec. An LLM agent reads the standardized cluster schema and writes a specification using the type-routed templates in Table 3. We use an LLM rather than rule-based extraction because reviews vary too much for fixed rules and the agent must reason across a 5-review cluster. We route by issue
255 type because a bug report needs reproduction steps, a feature request needs
260

acceptance criteria, and a performance complaint needs latency or memory numbers; a shared template would be incomplete or padded. The five templates are tracker-standard bug reports [5], Cohn user stories [40] with BDD acceptance criteria [41], ISO/IEC 25010 non-functional categories [42],
 265 Nielsen heuristics [43], and device-OS matrices [33]. The output passes through HITL-2, where a domain expert scores each spec on a five-dimension rubric (completeness, accuracy, actionability, specificity, clarity, each 1–5), following the LLM-as-judge calibration convention of Gilardi et al. [44], Pangakis et al. [45], and Zheng et al. [46]. Five dimensions rather than one,
 270 because a single quality number conflates completeness with clarity and the LLM cannot tell which one to fix on retry; any spec failing on any dimension returns for a second pass with dimension-level feedback.

SPECCOV faithfulness metric. We release SPECCOV, an extractive-coverage scorer that measures substantive-token overlap (length ≥ 5 , stop-word filtered) with quoted-string and coverage bonuses. Bands follow news-summarization conventions [47]; the metric targets the extractive-coverage sub-type of faithfulness [47, 48], a lexical correlate of factual consistency [49] (the remaining hallucination sub-types in Ji et al. [50] are out of scope). It is a lexical-grounding proxy, not a faithfulness classifier: Section 5.1 reports
 280 the experiment that withdraws it as one. We describe it here because it is part of the released pipeline and because the negative result only makes sense against its definition.

3.2. Using the IR in Response Generation

Stage 4 is a RAG layer over five fixed sources, 15,100 documents in the released ChromaDB index. Four are pre-indexed: app changelogs and re-
 285 lease notes (60 docs), FAQ and help documentation (40), 10,000 past review-response pairs from RRGGen [2], and 5,000 semantically similar past responses. The fifth is the validated IssueSpec from Stage 3, injected per query rather than pre-indexed. These five give the generator the answer, the voice, the
 290 closest precedent, and the structured intent without overlap.

3.3. Aligning the IR-Aware Generator

Stage 5 frames alignment as a CMDP [22, 20]: maximize a quality reward subject to a compliance constraint,

$$\max_{\theta} \mathbb{E}_{\pi_{\theta}}[R_{\text{quality}}] \quad \text{s.t.} \quad \mathbb{E}_{\pi_{\theta}}[C_{\text{compliance}}] \leq \tau,$$

with the Lagrangian relaxation $\mathcal{L}(\theta, \lambda) = \mathbb{E}_{\pi_\theta}[R - \lambda(C - \tau)]$ collapsing to
 295 single-objective optimization whenever the constraint is already satisfied.
 We use a CMDP rather than a scalar reward because a dev-rel response
 can be high-helpfulness and still out of compliance, like a polite promise the
 team cannot deliver, and a weighted sum hides exactly the trade-off the con-
 straint is meant to expose. The compliance scorer tracks four domain-specific
 300 dimensions absent from prior general-safety RLHF [20]: promise safety, infor-
 mation leakage, tone guidelines, and legal safety. The trainer is Constrained
 PPO [19, 20] with a KL-regularized reference policy; KTO [18] and DPO [17]
 are small-scale single-objective comparators bracketing the preference-data
 spectrum. Trainer specifics are in Appendix Appendix A. HITL-3 supplies
 305 two parallel feedback streams: *quality* feedback (helpfulness, specificity, em-
 pathy, accuracy, actionability) is the reward and *compliance* feedback is the
 constraint. Feedback also propagates backward, closing the Stage 5→Stage 1
 loop in Fig. 1.

4. Experimental Setup

310 4.1. Dataset, Deduplication, and Annotation Provenance

We use the RRGGen corpus [2]: 310,031 review-response pairs from 58
 Android applications. After exact-text deduplication that removes 94,448
 duplicate reviews (−30.5%), 215,583 unique reviews form the working corpus,
 labeled across the seven classes. Evaluation additionally uses a 490-review
 315 expert gold standard (70 per class × 7 classes) and 100 stratified clusters for
 Stage 3.

Annotation provenance. Of the working corpus, 79.49% of labels
 come from the V2 LLM annotator with no override, 18.08% are cleanlab-
 corrected against the verified anchor, and 2.43% are directly human-verified.
 320 The lead-author-versus-V2-LLM deviation is 25% on the 5,230-row anchor
 (1,307 deviations, 24.99%). We previously described this as replicated across
 two independent strata, the praise stratum and the anchor, on a combined
 10,271 rows. That was wrong: the praise stratum is 5,038 of the anchor’s
 5,230 rows, so it is 96% of the same sample rather than a second one, and
 325 the combined figure double-counted it. The rate is one measurement on one
 sample, and the praise stratum alone deviates at 25.84% (Table 4). This
 $\approx 25\%$ deviation rate, an order of magnitude higher than i.i.d. label-noise
 assumptions, motivates the confident-learning procedure of Stage 1.

4.2. Annotation Methodology

330 Five kinds of annotator contribute, following the multi-annotator convention of Gilardi et al. [44] and Pangakis et al. [45] who treat LLM annotators as independent raters for inter-rater statistics. **(i) The lead author** verified 5,230 reviews across minority strata, wrote 20 reference IssueSpecs, labeled the 490-review classification gold blinded to V2 LLM, and scored
335 the Stage 4 sets with method IDs hidden until aggregation. **(ii) An independent three-rater human panel** annotated the 490-review gold on the identical item set. **(iii) The same three people** formed the Stage 4 rater panel. **(iv) An LLM-as-judge** (Qwen2.5-3B-Instruct) scored IssueSpecs on the five-dimension rubric and sub-clusters on the Y/P/N purity
340 rubric [46], calibrated against the lead author. **(v) Corpus-labeling LLMs** (V2, cleanlab-V2, V5) are treated as independent raters for inter-rater statistics. Categories (iv) and (v) are model raters, not people; three humans contribute in total.

Label provenance, stated plainly. No LLM-generated label is used
345 as ground truth anywhere in this paper. LLM labels are the *object under correction*, not the yardstick. Every reported accuracy or κ is measured against a gold standard produced by human annotators. Where LLM raters do appear as raters, in the 99-review classification check, they are labelled as such and no headline claim rests on that panel.

350 **How the gold was constructed, and what that costs.** Annotators did not label freely. Each was shown the V5 classifier’s predicted label and asked whether it was correct, supplying a replacement when it was not, and the written protocol told them to accept a reasonable label and overturn only when confident. This makes annotation fast and agreement high, and both
355 effects flatter us. It also couples the yardstick to the system being measured: V5’s label is the starting point for every item, and V2 and the cleanlab-corrected labels never received that benefit of the doubt. The sample is stratified over V5’s predicted classes as well, which forces 70 items per class into a set where V2 predicts two of those classes almost never, so V2’s per-
360 class F1 of 0.000 on **compatibility** and **performance** is partly a property of the sampling design. The κ trajectory should be read as a comparison on a V5-seeded, class-balanced sample rather than as a neutral benchmark, and a free-labelling gold would be the way to settle it.

How rater conflicts are resolved. The 490-review gold is the majority
365 vote of three independent human annotators. Of the 489 usable items, 476 (97.3%) are unanimous and 13 (2.7%) are two-to-one splits resolved by the

majority; there are no three-way ties. One item was marked wrong by all three raters without a replacement label and is excluded rather than guessed. The per-item votes are released, so a reader can recompute any statistic
 370 on the unanimous subset alone. Before the main task all three annotators completed a shared 20-review calibration round whose disagreements were resolved against the written decision rules, which is where the systematic conflicts (slow versus broken, device-specific versus general) were settled.

A disclosure about an earlier annotation round. The 490-review
 375 gold was annotated twice. In the first round three sheets were distributed, and the two returned alongside the lead author’s carried annotation columns identical to hers cell for cell, so no genuine agreement could be computed from them and they are not used anywhere in this paper. The round we report is the second, and the agreement statistics come only from it. We
 380 disclose this for the same reason we disclose the Stage-4 case in Section 5.2: a paper that offers a check for non-independent ratings should say where its own artifact failed that check. Both rounds are released.

Stage 4 was likewise rated twice. The first round scored a 400-row set whose conditions were produced by template composers; we discarded it,
 385 for reasons given in Section 5.2. The round we report is a 200-row set in which both arms come from the same model and differ only in whether the IssueSpec was supplied, all 200 rows scored independently by all three raters, blind to condition. Ratings are persisted as spreadsheets with JSON master keys, blinding shuffles conditions per pair, and the assignment unseals only
 390 at aggregation. Table 4 consolidates agreement and disagreement across all pairs.

Two distinct three-rater statistics in Table 4 should not be conflated. The classification gold uses three independent *human* raters and reaches Fleiss $\kappa = 0.968$ on the Y/N verdict and 0.979 on the seven-class label. The
 395 $\alpha = 0.285$ row is a separate check on 99 reviews that uses the lead author plus two Qwen2.5-3B prompts (concise and chain-of-thought) as raters, measuring seven-class classification agreement rather than cluster purity. We reported this row as $\alpha = 0.451$ and as a purity check; both were wrong. The value came from an implementation that credited each rating with a coincidence
 400 against itself, inflating observed agreement; the corrected figure agrees with the reference `krippendorff` package to four decimals. The three Cohen’s κ in the same table were computed with scikit-learn and are unaffected. The correction moves the row from the moderate band into the fair band, which weakens the LLM-rater panel rather than any claim we make.

4.3. Experimental Design

A single-LLM headline raises a structural-bias concern, so we replicate Stage 3 across four LLMs from three families, Anthropic Claude (Opus 4.7), Meta Llama-3.3-70B-Instruct (via Groq), and Alibaba Qwen2.5-{3B, 1.5B}-Instruct, and check that cross-family rank order is preserved (Table 6). We pre-register three experiments, one per research question.

Experiment 1 (RQ1, translation quality). Five conditions on the same 100 stratified clusters: (a) LLM with taxonomy grounding (the full Stage 3 system), (b) LLM without taxonomy grounding (free-form), (c) raw review summary (a lower bound concatenating the cluster’s reviews), (d) lead-author reference, and (e) human-written GitHub issues mined from three open-source Android projects (the practitioner-quality upper bound). Each spec is scored on the five-dimension rubric, with significance by paired Wilcoxon signed-rank tests.

Experiment 2 (RQ2), as reported. The design we report holds the generator fixed. The same model, given the same retrieved context, writes two replies for each of 100 reviews, differing only in whether the validated IssueSpec is in the prompt. All 200 rows are rated blind by three raters on quality, specificity, and helpfulness, and analysed with a paired Wilcoxon test and a paired bootstrap interval. *The discarded design*, described because the paper reports its withdrawal rather than because it supports a claim, used the same 100 clusters to drive four conditions produced by template composers rather than by a model: (a) an rrgen-style baseline (review only), (b) a prompt baseline (CoRe-style dev-rel prompt with lightweight entity extraction), (c) RAG without IssueSpec, and (d) the full system. It was evaluated with automatic metrics (BLEU, ROUGE-L, BERTScore) and 400 blinded paired human ratings, with significance by paired Wilcoxon, Friedman with Nemenyi post-hoc [52], Bradley-Terry, and McNemar at Bonferroni-corrected $\alpha = 0.05$, and effect sizes following [53, 54]. The headline comparison runs single-shot vanilla RAG [10] rather than an agentic loop, so that the IssueSpec’s contribution is not confounded with orchestration; the agentic comparison against Self-RAG [28] and DSP [29] runs separately on a 10-review feasibility sample and is reported as a demonstration rather than a powered comparison.

Experiment 3 (RQ3, dual- vs. single-objective RLHF). Five policies run on a distilGPT2 proof-of-concept (82M parameters, 400 SFT samples, 296 KTO pairs, 100 DPO pairs): SFT-base, single-objective KTO [18], single-objective DPO [17], a constrained-proxy variant, and dual-objective

Lagrangian Constrained PPO [19, 20]. We measure head-to-head BLEU, ROUGE-L, and BERTScore on a 100-review test set, plus the rule-based safety-violation rate.

5. Results

How to read this section. Each research question owns one subsection. Section 5.1 answers RQ1, is the typed IR producible at scale, with Table 8 and Figure 2. Section 5.2 answers RQ2, is the IR load-bearing downstream, with Table 10; the answer is no, and the subsection explains what our earlier positive answer was measuring. Section 5.3 answers RQ3 and reports a negative result. Sections 5.4 and 5.5 test whether individual components earn their place, and Section 6.1 states which ones do. Table 1 summarises all of it in advance.

5.1. Is the IR Producible at Scale?

Template-fill measures schema conformance, and its limit is worth stating exactly. The criterion is the templates’ own required fields, and the free-form condition fills the four shared fields and none of the type-specific ones, so its score is determined entirely by the issue-type mix: $(30 \cdot 4/7 + 30 \cdot 4/6 + 40 \cdot 4/5)/100 = 0.691$, with zero variance across specs. The 0.69 is therefore a constant, not a measurement, and the 0.96-versus-0.69 gap says that routing produces type-specific fields at all. It carries no information about whether the routed text is better written. Scoring the same conditions on bare field presence instead of the strict criteria gives 1.000 for the routed condition against 0.678 for free-form, versus 0.959 and 0.678 under the strict criteria: for the free-form arm the content thresholds remove exactly nothing, so the whole reported gap is schema conformance. And a hand-written bug specification whose description is the word “something” repeated thirty-five times, with three five-word steps built from the criterion’s own action-verb list, passes all seven fields and scores 1.000, above our headline. A reader should treat 0.96 as schema conformance and nothing more. Under strict content-validity criteria (at least three reproduction steps with action verbs, descriptions ≥ 30 words, and a full As-a/I-want/So-that triple for user stories), the LLM-with-taxonomy condition achieves a template-fill rate of 0.96 (Table 8). Free-form LLM reaches 0.69, the same as the lead-author reference set ($n = 20$). Raw concatenation lands at 0.34, and real GitHub issues mined from three open-source Android projects reach 0.53. The LLM-vs-GitHub

gap reflects coverage rather than content quality: the LLM is prompted to fill every field, whereas humans leave optional fields blank. A very-strict sensitivity sweep preserves the ordering, Claude dropping from 0.97 to 0.71 with the Claude-vs-GitHub gap shifting only -0.085 .

SPECCOV scores, and a correction to how they were computed. An earlier version of our scorer applied per-condition floors, raising `raw_summary` to 5 and the human reference to 4 regardless of content. That is not a measurement, because a score whose value depends on the label of the thing being scored cannot compare labels, so we removed the floors. Across 320 specs the LLM-with-taxonomy condition averages 4.19, free-form LLM 3.38, raw concatenation 4.47, and the human reference 3.45. The two LLM conditions are unchanged, because no floor was ever applied to them; the floors had inflated the two comparison conditions, so removing them makes our own system look better rather than worse, and we note that so the direction of the correction is on the record. Read on their own these numbers rank raw concatenation highest at 4.47, the trivial upper bound obtained by copying the source, with the taxonomy condition second at 4.19. Section 5.1 shows they should not be read on their own. Top-ranked aspects by PageRank centrality are `ad` (0.040), `phone` (0.011), and `battery` (0.007).

SpecCov does not measure faithfulness

An extractive-coverage score is only useful if it agrees with what a person calls faithful. Three human raters independently scored 60 specs, 20 from each of three generators, shuffled and unlabelled, on a 1–5 faithfulness scale, and separately listed every concrete claim the source reviews do not support. One qualification is needed first. One rater completed the corrected sheet, which shows exactly the evidence SPECCOV itself scores against; the other two completed an earlier draft showing a subset of it, so they judged against less evidence than the metric uses and their flags are biased toward over-flagging. We therefore lead with the rater who saw the full evidence. The metric fails all three checks on either reading.

It does not rank specs the way humans do. Spearman ρ between SPECCOV and human faithfulness is $+0.045$ ($p = 0.73$), $+0.015$ ($p = 0.91$), and $+0.226$ ($p = 0.08$) for the three raters, and $+0.147$ ($p = 0.26$) against their mean. No correlation reaches significance.

It does not detect the claims humans call unsupported. On the 60 specs, one rater marked 45 as containing at least one unsupported detail and 15 as

515 clean. SPECCOV scores those two groups at 3.64 and 3.93, a difference in the right direction but far from significant (Mann-Whitney $p = 0.34$). A metric that detected hallucination would separate these groups.

It orders the generators backwards. SPECCOV ranks the taxonomy condition (4.20 on this subsample) above the human reference (3.45); the humans 520 rank them the other way, human reference 4.10 and taxonomy 3.50. The mechanism is visible in the definition: templated specs reuse source vocabulary densely, which the score rewards, while a human writer condenses and paraphrases, which the score penalizes even when nothing is invented.

We therefore withdraw SPECCOV as a faithfulness measure. Lexical 525 grounding is not a usable proxy for faithfulness on this task, and the automatic 320-spec rubric that appeared to support it is not independent evidence, since its faithfulness dimension is computed by the same overlap procedure. Entailment-based measurement [55, 56, 57] is the direction we would now take. We release the scorer, the 60-spec human ratings, and this 530 analysis, because a documented negative result on a plausible-looking metric is more useful to the next person than a metric that was never tested.

Cross-LLM replication (bias control). Llama-3.3-70B reaches 0.80 template-fill, Qwen2.5-3B 0.74, and Qwen2.5-1.5B 0.48 under the same strict criteria; the templated > free-form > raw ordering holds across all four 535 models (Table 6, with the two caveats its caption records). On two per-class structural-compliance checks, Llama-3.3-70B attains complete template compliance on the stratified sample: all 29 bug specs include three or more reproduction steps with action verbs (vs. Claude’s 73%), and all 30 feature specs include the As-a/I-want/So-that triple (vs. Claude’s 97%). These mea- 540 sure structural compliance, not content correctness. The bias control that matters is cross-family rank preservation: the IssueSpec contract transports across families rather than depending on a single proprietary generator.

Cross-family rubric quality. Template-fill says nothing about whether the content is any good, so we ran the same Qwen2.5-3B judge and five- 545 dimension rubric over the other generators’ specs on matched clusters (Table 5). On the 14 clusters all four generators cover, mean rubric score follows the same order as template-fill: Claude 4.09, Llama-3.3-70B 3.96, Qwen2.5-3B 3.83, Qwen2.5-1.5B 3.77. We state plainly what that ordering is worth. At $n = 14$, or $n = 13$ for the three pairs involving Qwen2.5-1.5B, none of the 550 six pairwise differences reaches significance (Wilcoxon p between 0.18 and 0.75), and the adjacent gaps are the same size as the run-to-run generation noise measured in Section 6.4. The four-way ordering is therefore consistent

with the template-fill result rather than independent confirmation of it. The Claude-versus-Llama comparison in Panel B, which has twice the sample and spans all five issue types, reaches a nominal +0.38 ($p = 0.015$). We previously asserted this as the one cross-family quality difference we stand behind. We withdraw that assertion. It belongs to a family of seven Wilcoxon tests reported in the same tables with the same judge and rubric, and Bonferroni or Holm correction over that family puts it at $p = 0.104$; an exact sign test on the same 28 pairs gives $p = 0.078$. The difference is descriptive, and we no longer claim a cross-family quality difference at all. The spread is much narrower than the (unmatched) template-fill spread, which is informative in itself: the capability gap between these models shows up mostly as whether fields get filled at all, not as the quality of the text once they are. Completeness is the dimension that separates them (4.00 down to 3.00), while accuracy is nearly flat (4.50 for three of the four). Two caveats belong with these numbers. The four-way panel is bug-report clusters only, because that is where the two Qwen runs overlap, and the judge is itself Qwen2.5-3B-Instruct, so the Qwen2.5-3B row carries a self-preference risk that the others do not.

Five-Dimension Rubric. The 5-dimension Likert rubric [44, 45, 46] is operationalized with Qwen2.5-3B-Instruct as judge over the complete set: all 100 taxonomy specs and all 20 lead-author reference specs in a single run (Table 7). The taxonomy condition reaches 3.94/5 (sd 0.59), above the hypothesized lower bound of 3.5. An earlier stratified 28-spec subsample under the same judge gave 3.89, so subsampling cost 0.05 of a Likert point, which bounds how misleading the small-sample version could have been. Per issue type the taxonomy condition ranges from 4.23 on performance and 4.02 on bug reports down to 3.45 on usability. The reference set is a judge calibration, not a competitor: it scores 3.45 overall, with the gap concentrated in actionability (2.85 versus 3.86), because the lead author filled title, description, severity, and component but left template-specific fields empty, writing a reference spec by hand being a different task from filling a routed template. The comparison therefore says that a human writing freely produces less tracker-ready output than a type-routed generator. It is not evidence that the specs are better than expert judgement on content, and we do not read it that way.

Stage 1 Classification Recovery. The Cohen’s κ trajectory against the 490-review expert gold crosses three Landis-Koch [51] agreement bands (Figure 2): the V2 LLM achieves $\kappa = 0.163$ (slight); after cleanlab correction [21] $\kappa = 0.333$ (fair); the V5 classifier retrained on the corrected labels

reaches $\kappa = 0.592$ (moderate). The trajectory does not depend on any single annotator: scored against the majority-vote gold of the three-rater human panel ($n = 489$ after excluding one review left without a correction label), the same three classifiers reach $\kappa = 0.165$, 0.334 , and 0.590 , within 0.002 at every step. On the held-out 307-review subset (183 of the 490 entered training, mostly through the stratified cap rather than the anchor proper), V5 reaches $\kappa = 0.616$. We previously wrote that this is “slightly above the 0.592 on the full 490, ruling out a contamination artifact.” Two audits make that untenable as written.

First, “held out” is not exactly true. All 100 keyword-mined **compatibility** rows in the training augmentation are verbatim copies of reviews already in the corpus, appended past index 215,583 with the label changed. Eight of them duplicate a held-out review, seven with the same label the expert assigned, and the index-based replay cannot see this because it filters on position rather than text. Dropping those eight gives $\kappa = 0.609$ on $n = 298$.

Second, and more seriously, the split is a replay rather than a recorded fact, and 0.616 is the most favourable replay. The reconstruction shuffles each class with seed 42 and iterates classes in Python dictionary insertion order. Iterating them in sorted order instead, changing nothing else, gives $\kappa = 0.594$; seeds 0, 1, 2, 7, 123 and 2024 give 0.602 , 0.605 , 0.617 , 0.590 , 0.583 and 0.606 . The published figure is at the top of that range, and the direction of the inequality against 0.592 flips under most alternatives. What survives is the weaker statement: held-out κ is indistinguishable from full-set κ , across replays and after removing the duplicated rows. That still argues against memorisation, which is what the sentence was for, but it does not do so by being larger, and we should not have leaned on a direction this fragile.

The per-class recovery story (Table 9) makes the structural collapse visible. On the two minority classes the V2 LLM cannot see at all, F1 jumps to 0.83 for **compatibility** and 0.77 for **performance**. Applied across the full corpus, V5 agrees with the cleanlab correction on 88.66% of the 40,291 corrected cases. V5 was trained on a corpus in which 38,985 of those corrected rows carry the corrected label, so its agreement is substantially a measure of fit to its own training signal, not an independent second opinion. We report it as a consistency check and do not claim the two signals are independent.

The stronger evidence is the held-out gold. V5’s training corpus is built with a stratified per-class cap of 15,000 rows at seed 42; replaying that cap identifies which corpus rows survived into training, 183 of the 490 gold re-

views, leaving 307 capped out, of which 306 carry a gold label and are
630 scorable. The split is an artifact of class-balancing rather than of anchor
membership, but it is not unbiased in the way we first described. The cap
binds only on the four classes V5 predicts most often, so membership of the
unseen set is decided by V5’s *predicted* class rather than at random. Scored
by expert label, the unseen 307 still spans all seven classes and includes 38
635 **compatibility**, 32 **usability**, and 31 **performance** reviews, so the held-out
comparison does cover the classes carrying the recovery. What it cannot rule
out is a subtler selection effect: reviews V5 predicted confidently into large
classes are over-represented among the unseen, and those may be easier. As
the replay analysis above shows, the ordering of 0.616 against 0.592 is not sta-
640 ble, so the defensible reading is that the two are indistinguishable; a model
that had merely memorised its training labels would fall on unseen items
rather than hold at the same level. The split is reconstructed by replaying
the cap rather than read from a saved manifest, so it is a faithful recon-
struction rather than a recorded fact, and we release the replay script. The
645 choice of confident-learning thresholds is not load-bearing: a full 3×3 grid
over (`anchor_conf`, `max_llm_prob`) $\in \{0.65, 0.70, 0.75\} \times \{0.15, 0.20, 0.25\}$
keeps V5 endorsement in an 82.7–91.0% band, with the headline (0.70, 0.20)
at 88.66% near the grid center.

To rule out memorization of the 200 synthetic compatibility templates
650 used during V5 training, we evaluate V5 against Maalej’s 5,008 labels under
a label-space cross-protocol test. Review texts appear in the anchor under the
seven-class taxonomy, but predictions are scored against Maalej’s separate
taxonomy, which contains no **compatibility** class, making any compatibil-
ity prediction out-of-distribution with respect to the evaluation labels. On
655 the five overlapping classes ($n = 4,560$), V5 achieves macro $F1 = 0.676$,
weighted $F1 = 0.730$, and accuracy = 72.0%. V5 also flags 432 reviews
(8.6%) as **compatibility**; the top-twenty-confidence flags read as canon-
ical compatibility complaints (“crashes on iphone 5 and ipad 3”), all labeled
bug_report in Maalej because Maalej offers no alternative. On this out-
660 of-distribution evaluation the detector performs concept-recognition rather
than template-matching.

A caveat about the class itself, found by auditing the training file. Of the
310 **compatibility** rows V5 trains on, 300 are augmentation: 200 template-
generated and 100 keyword-mined. The corpus is lowercased and lemmatised
665 throughout, and we verified that not one of its 215,583 rows contains a single
uppercase character, whereas 194 of the 200 synthetic rows do. V5 is a cased

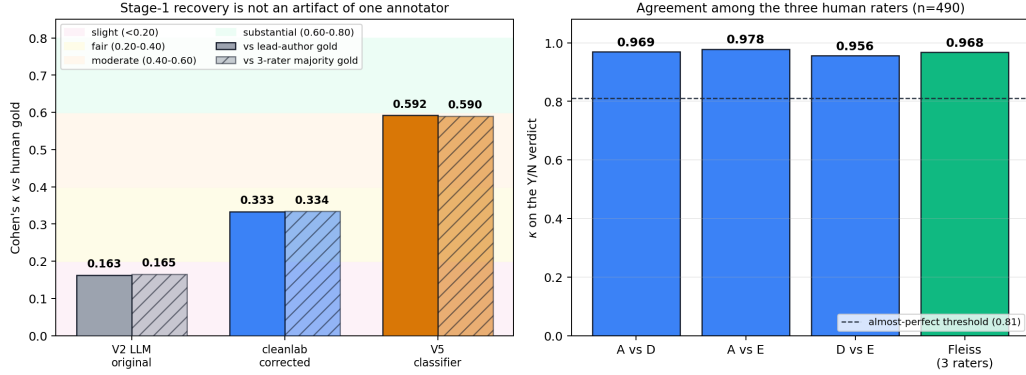


Figure 2: Stage-1 classifier recovery and the gold standard it is measured against. *Left*: Cohen’s κ for the three classifier versions, plotted twice, once against the lead-author gold (solid) and once against the three-rater majority-vote gold (hatched). The two are within 0.002 at every step, so the 0.163 to 0.592 trajectory does not depend on any single annotator. Background bands are the Landis-Koch [51] agreement ranges. *Right*: agreement among the three human raters on the same 490 reviews, pairwise and as a three-rater Fleiss κ , all above the almost-perfect threshold. Read the left panel for the result and the right panel for the reliability of the yardstick it uses.

RoBERTa, so any capital letter is a free and perfect marker for “synthetic compatibility”. The internal test F1 of 0.74 for this class is therefore not a meaningful number and we withdraw it. The gold-standard compatibility F1 of 0.83 is not affected by the cue, because the gold is lowercase corpus text where it never fires; it is carried instead by the 100 mined rows, which are the verbatim duplicates described above. The cross-protocol result stands, since Maalej’s text is also not drawn from the synthetic set, but a reader should treat every compatibility number in this paper as resting on 300 augmented rows rather than on natural corpus evidence.

5.2. Is the IR Load-Bearing? A Result We Had to Withdraw

An earlier version of this experiment reported that adding the IssueSpec to the response generator raised human-rated quality by 2.35 Likert points. That number was wrong, not in its arithmetic but in what it measured, and the correction is the most important thing this section has to report.

What was wrong. The four conditions in that experiment were not produced by a language model. Each was produced by a deterministic template composer: a Python script that selects sentence fragments from a hand-written bank and fills slots, with no language model in the loop. The com-

685 posers differed in authorial register as much as in what information they were given. The `no_spec` arm emitted lemmatised, ungrammatical text averaging 78 words (“we be sorry that the login loop you describe have got in the way”); the `full` arm emitted polished prose averaging 123 words, with cleanup passes and an empathy phrasebook that no other arm received. A
 690 rater comparing them was separating two writing styles, and the presence or absence of the IssueSpec was confounded with that separation.

The clean comparison, and what we can and cannot verify about it. We re-ran the experiment with the generator held fixed. Both arms come from the same model on the same 100 reviews, and the intended difference
 695 is whether the validated IssueSpec appears in the prompt. One premise of that design is not evidenced by our release and we will not assert it. Neither response file stores the prompt or the retrieved text, and the two files record their retrieval sources under different label schemes with different counts, so on the released artifact no row has an identical recorded source list. Both
 700 arms draw from the same candidate pool, and that much is checkable, but whether the retrieved context was identical per review is not recoverable. A reader should therefore treat the comparison as holding the generator and the review set fixed, not as a strict single-variable manipulation. Retrieval differences would add noise rather than a systematic advantage to either arm,
 705 which makes a null easier to obtain, so this does not rescue the withdrawn positive result; it does mean the null is weaker evidence than a fully controlled design would give. The two arms average 105 and 97 words with distinct-1 at 0.736 and 0.760, so they are close on the surface features a rater could use to tell them apart, though the length difference is itself significant (paired
 710 Wilcoxon $p = 6 \times 10^{-8}$).

The IssueSpec does not measurably improve response quality. Pooled over the three raters the gain is +0.04 Likert points, 95% bootstrap CI $[-0.10, +0.18]$, paired Wilcoxon $p = 0.54$ on 100 pairs, Cliff’s $\delta = 0.04$. Each review’s two replies are shown in a randomised order; the slot assignment
 715 came out 64/36 rather than balanced and raters show a small preference for the first slot, so we also computed a slot-stratified estimate, which moves the pooled figure to +0.03 and each rater’s by at most 0.03. We use the unadjusted numbers throughout so that every figure in Table 10 is on one estimator, and release the stratified values with the data. Specificity behaves
 720 the same way, pooled -0.00 ($p = 0.87$), and the helpful rate is flat at 94% in both arms. On the evidence of this design we cannot claim that supplying the typed specification improves the response a user receives. We should

be equally careful in the other direction. With 100 pairs and the observed spread, the smallest effect this design could detect at 80% power is about
725 a fifth of a Likert point, so what we have ruled out is a large effect, not a small one; detecting a gain the size of the one we measured would need on the order of 2,500 pairs. The honest statement is that the effect, if any, is small enough that this study cannot see it, not that it is zero.

The raters disagree with each other, and that is the interesting
730 **part.** Table 10 breaks the result out. The lead author, who designed the system, prefers the IssueSpec arm decisively: $+0.56$ quality ($p = 2 \times 10^{-11}$, 59 wins, 5 losses, 36 ties) and $+0.39$ specificity ($p < 10^{-7}$). The two independent raters lean the other way, -0.29 and -0.15 , neither significant at $\alpha = 0.05$ under the zero-handling scipy applies by default ($p = 0.06$ and 0.09 ;
735 the first rater has 45 tied rows of 100, and under Pratt’s method that becomes $p = 0.044$). That rater’s bootstrap interval, $[-0.56, -0.01]$, excludes zero, so the test and the interval disagree at the boundary. We state both rather than pick the one that reads better: on this rater the evidence sits exactly on $\alpha = 0.05$ and moves either side of it depending on defensible choices.
740 That is itself the point of the section, since the disagreement between raters is larger than the effect under measurement. Inter-rater reliability is correspondingly low, quadratic-weighted κ of 0.03, 0.04 and 0.50, against 0.97 in the discarded template-based round. High agreement there was a symptom of the confound: when one arm is visibly broken, raters agree easily. Here
745 they do not.

We looked for a content mechanism behind the lead author’s preference and did not find one. Correlating each rater’s per-review preference against the spec-derived features the IssueSpec should introduce, named devices, reproduction steps, and component mentions, produces no correlation that
750 survives inspection for any rater; the largest is $\rho = 0.19$ ($p = 0.06$) between the lead author’s preference and reply length. So we can describe the disagreement but not explain it. Two readings remain open. The lead author may be perceiving triage value that a general reader has no reason to weigh, in which case the right evaluation asks developers rather than readers. Or
755 the preference may be expectancy on the part of the person who built the system. This design cannot separate those, and we do not assert either.

Why we believe the raters this time, and what the check does and does not show. In the discarded round the two additional raters deviated from the lead author on only 16 and 19 of 400 rows, those deviation
760 sets did not overlap on a single row, and every deviation was exactly one

point. We initially read the zero overlap as the damning fact. It is not: with deviations that rare, the hypergeometric probability of zero overlap under perfect independence is 0.45, so zero overlap is the single most likely outcome and carries almost no information. The discriminating observation
765 is the second one. On the quality column every deviation in both sheets had the same magnitude, one point, which is what a vector perturbed by a fixed step produces and not what human disagreement produces. We state the column because the specificity column in the same sheets does show magnitudes of one, two and three, so the signature is present in one column and not the other. In the reported round the raters differ from the lead
770 author on 133 and 145 of 200 rows, the deviation sets overlap on 101 where chance predicts about 96, and the deviations take four distinct magnitudes from one to four points. We release the check with the data, and we state its weak prong plainly, because a check offered as a contribution should come with its own limits attached.
775

What survives. Less than we first thought. The claim that the typed IR carries downstream response quality does not survive, and neither does the retrieval finding we had treated as the surviving half. We first reported that retrieval alone was indistinguishable from a no-context baseline, then
780 corrected that to significantly worse; both statements compared the retrieval arm against the *prompt* baseline, which is not a no-context condition because it carries a developer-relations instruction. Against the arm that sees the review text only, retrieval scores 2.24 against 2.31, a difference of -0.07 at $p = 0.90$, and all three arms come from the discarded template-composer round
785 in any case, so none of them supports a claim in either direction. Stage 4 leaves us with a null and nothing else. Stage 3’s contribution, producing tracker-ready structured records at corpus scale, stands on its own evidence in Section 5.1; it no longer has a demonstrated downstream payoff.

5.3. Aligning the IR-Aware Generator at Scale

Five policies were trained on distilGPT2 (settings in Appendix Ap-
790 pendix A). Head-to-head on a 100-review test set, per-policy BLEU-1 / ROUGE-L / BERTScore F1 are: SFT-base 0.090 / 0.097 / 0.802; KTO 0.068 / 0.074 / 0.792; DPO 0.084 / 0.088 / 0.800; **constrained-proxy 0.137 / 0.131 / 0.806**; Lagrangian PPO 0.087 / 0.091 / 0.798. The
795 constrained proxy leads three of four metrics, with +53% BLEU-1 over SFT-base; the rule-based safety-violation rate is uninformative here because compliance is satisfied at initialization.

The dual-objective claim is not demonstrated, and we report that as the result. At distilGPT2 scale the compliance scorer is satisfied at initialization ($C = 0.94 \geq \tau$), so λ collapses to zero and the update reduces to single-objective optimization. Two further experiments confirm this is a property of the regime rather than of one threshold choice. A re-run replacing the permissive scorer with the four operational compliance checks and tightening τ from 0.50 to 0.90 records 0 binding steps out of 30 and a single violation across the whole run; a sweep over $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$ leaves the pass rate at 100% for four of the five policies at every threshold, with only SFT-base dipping, to 99% at $\tau \geq 0.9$. A model of this size cannot plausibly violate the rubric, so the CMDP machinery is untested under binding conditions. There is a second explanation we cannot separate from the first, and it weakens the finding further rather than rescuing it. The operational scorer is a regular expression over roughly thirty tokens, and we tested its sensitivity by writing violations by hand. Nine replies that plainly breach the four rubric dimensions, phrased away from the patterns (“Our engineers have committed to a permanent resolution by Friday”, “Ticket ENG-4412 in our tracker covers this regression”), all score a perfect 1.00. Empty output, whitespace, and gibberish also score 1.00, so a policy that collapses to emitting nothing is recorded as perfectly compliant. The quality scorer has the mirror defect: a bare list of eleven rubric keywords scores 1.000 against 0.672 for a genuine reply. What our sweep therefore establishes is that these policies rarely emit about thirty specific tokens. Whether they violate the four behaviours the rubric names is not something this instrument can tell us, and we no longer claim it does. Every compliance number in this section should be read as a property of the detector, and the +53% BLEU-1 figure is a single-objective result, not evidence of a quality-versus-compliance trade-off.

The constraint binds once the policy can violate it, as this instrument measures violation. We originally read the obstacle as capability rather than parameter count: distilGPT2 never produces the fluent phrasing the rubric penalises, so it cannot violate and the constraint has nothing to hold back. Given the detector’s measured insensitivity that reading is not the only one, and we cannot distinguish a policy that does not violate from a policy whose violations this scorer does not see; every violation rate below should be read with that ambiguity attached. We first justified the capability reading by observing that 5.2% of our own Stage-4 generations trip the scorer. That figure is from the four withdrawn template-composed arms (21 violations in 400 replies). Scored the same way, the two model-generated arms

we actually defend violate on 4 of 200 replies, 2.0%, and one of the four is an information-leak rather than an over-promise. The zero-shot Qwen2.5-1.5B probe gives 2 violations in 40 replies, 5.0%, whose 95% interval runs from roughly 1% to 17% and therefore cannot be distinguished from 2.0% either way. distilGPT2 gives roughly zero. The honest version of the argument is weaker than the one we first made: a 1.5B instruct model violates the rubric occasionally where distilGPT2 essentially never does, and that is enough for the constraint to bind, but we cannot claim it matches the rate of our own generations. We re-ran the Lagrangian loop with Qwen2.5-1.5B-Instruct as the policy, LoRA adapters over the attention projections, the frozen base as KL reference, and $\tau = 1.0$, everything else unchanged. The behaviour inverts. Over 90 steps the constraint binds on 13 (14.4%), and λ leaves zero, rises monotonically to 0.65, and never collapses back; a 30-step run reaches $\lambda = 0.125$, so the multiplier scales with the horizon rather than saturating. Split by thirds: 1 binding step and $\lambda = 0.05$, then 7 and 0.40, then 5 and 0.65. The qualitative claim that the CMDP degenerates to single-objective optimization is therefore a property of the proof-of-concept policy, not of the formulation.

What is and is not shown. The two series move in opposite directions across the run, quality from 0.565 to 0.590 and compliance from 0.997 to 0.989, while λ climbs. We are careful not to call this a trade-off: the step-level correlation between quality and compliance is $\rho = -0.12$ ($p = 0.25$), so what we have is two separate trends over a single run and no measured coupling. What we cannot show is the multiplier growing large enough to pull the policy back into compliance; at a 5% base violation rate and a batch of four, roughly one violating sample arrives every seven steps, and 90 LoRA steps do not accumulate enough constraint pressure to reverse the drift. So the correct statement is narrower than a confirmed RQ3 and wider than the earlier null: the dual objective engages, the multiplier responds monotonically to the tokens it matches, the tension between the objectives is not measurable on our data, and closing the loop back to compliance needs a longer horizon or a higher base violation rate than a 1.5B instruct model supplies. What this layer contributes is the CMDP formulation instantiated over four dev-rel constraints, an end-to-end trainer that runs, the threshold sweep, the capability probe, the binding-constraint re-run, and the released testbed.

5.4. Cluster Quality and Purity

Stage 2 was redesigned from a flat UMAP+HDBSCAN baseline (194 clusters, mean size 471.0) to the three-layer KG-hierarchical design (605 sub-clusters, mean size 15.9). The two were not run over the same reviews: the flat baseline clusters 91,368 reviews (117,958 in, 26,590 assigned to HDBSCAN noise) drawn from the full relabelled corpus, whereas the KG-hierarchical design is built on the 8,404-review knowledge-graph sample, of which 5,689 distinct reviews land in a sub-cluster. A sixteen-fold difference in the underlying review set is a third reason, on top of the two below, why the head-to-head intrinsic comparison is not a comparison. We compute two complementary validity axes: *intrinsic geometry*, Davies-Bouldin [58], Calinski-Harabasz [59], silhouette [60], following standard practice [61, 62, 63]; and *extrinsic cluster purity* [64, 61] under a Y/P/N weighted-purity rubric, $\text{purity}_w = (|Y| + 0.5|P|) / (|Y| + |P| + |N|)$, with Y = all 5 share theme, P = 3–4 of 5 share, N = fewer than 3 share.

We reported that KG-hierarchical achieves $5.4\times$ lower Davies-Bouldin and $1.9\times$ higher Calinski-Harabasz than the flat baseline. We withdraw those two figures. The script that produced the flat baseline’s intrinsic metrics draws its cluster labels as a random subsample of 10,000 reviews spanning all 194 clusters, but collects the review texts to embed by walking the clusters in order, and the first flat cluster alone contains 10,507 reviews. The embeddings therefore come from a single cluster while the labels span 194, so the flat column’s DB of 12.15 and CH of 0.98 measure random labelling rather than the flat partition. The KG column was computed on a correctly paired set, so the comparison was a valid measurement against a broken one. Table 11 keeps the KG column and marks the flat intrinsic column as withdrawn. Repairing the alignment would not make the two columns comparable in any case: the flat side embeds 10,000 full reviews while the KG side embeds three representative reviews per cluster, so the two are computed over different kinds of object. The count-matched A1b ablation, which embeds representatives on both sides, is the only intrinsic comparison we report, and it is unaffected. Extrinsic purity is also unaffected: the KG’s finer clusters score 0.39 against the flat baseline’s 0.625 under the same Qwen judge, and the flat baseline’s 0.66 lead-author purity sits in the substantial-agreement band (≥ 0.61) against KG-605’s fair band (≥ 0.21) on the Landis-Koch [51] convention adapted to weighted purity, reflecting the granularity-versus-coherence trade-off A1b makes explicit. The KG loses on four of five issue types on per-type weighted purity.

910 **Where the KG actually wins.** On the prioritization axis the story reverses. Of the 10,534 aspect nodes in the KG, the top five by PageRank are touched by 55% of reviews and the top ten by 66%. We give the deduplicated figures; counting a review once per aspect it mentions inflates these to 74% and 96%, and that inflated pair should not be used. A developer triaging in
915 PageRank order therefore reaches 55% of the complaint volume after reading five aspect buckets and 66% after ten. We previously wrote “three-quarters” in this sentence, which was the discarded 74% reappearing two lines after we told the reader not to use it. We also previously claimed a “roughly $10^3\times$ concentration ratio” against random aspect ordering. We withdraw that too:
920 we never computed it, and random ordering is the wrong baseline in any case, since the operative comparison is against ordering aspects by raw frequency, which we did not run. The KG additionally enables per-aspect drill-down, distinguishing `battery`→Samsung-drain from `battery`→fast-charge.

5.5. Ablations

925 An ablation removes one component and re-measures. Eight are listed, counting A1b, which a reviewer asked for; seven were run and one, A7, is deferred. Three came back against the design, and Table 1 carries the verdicts.

A1, remove the knowledge graph and feed flat clusters into
930 **Stage 3.** Stage 3 quality does not change, because the translator consumes a cluster centroid and its representative reviews, not the graph edges. The KG is therefore not a Stage 3 dependency, which narrows where its value could lie.

A1b, compare the KG against flat clustering matched to the
935 **same number of clusters.** The KG produces 605 sub-clusters and the flat baseline 194, and intrinsic metrics move with cluster count on their own, so any comparison between the two conflates structure with granularity. Re-running agglomerative clustering forced to exactly 605 clusters, on 1,815 representative reviews (3 per sub-cluster), removes the confound. The
940 flat version then wins on all three metrics, Davies-Bouldin 1.12 against 2.24 (lower is better), Calinski-Harabasz 4.21 against 1.85 (higher is better), and silhouette +0.148 against −0.234 (a negative value means the average point sits closer to a neighbouring cluster than to its own). The honest conclusion is that the KG does not produce geometrically better clusters. What it
945 produces is an ordering, and prioritization is the only cluster-layer claim we make.

A2, remove hierarchical sub-clustering. Weighted Y/P/N purity, judged on five representative reviews per cluster, is 0.625 for the flat 194 and 0.390 for the KG’s 605. Finer granularity costs coherence here rather than buying it. Taken with A1b this is the second result against the cluster layer, and we report both.

A3, remove taxonomy grounding. The LLM writes a free-form issue description instead of filling a routed template. Strict template-fill falls from 0.96 to 0.69, and the per-type structural columns fall to exactly zero, because an unrouted generator never emits a reproduction-step list or an As-a/I-want/So-that triple at all. This is the ablation that isolates the paper’s central mechanism. The gain is not the language model, it is the routing.

A4, remove the HITL-2 spec review. Benchmark scores do not move, and the reason is procedural rather than substantive. HITL-2 is a production-time gate; the 100 specs used in the evaluation were scored before any human correction, so the headline numbers already describe pre-HITL output. The ablation therefore says nothing about whether HITL-2 helps in deployment, and we do not claim that it does.

A5, remove retrieval and keep the IssueSpec. BLEU-1 moves by -0.006 , ROUGE-L by -0.008 , and BERTScore F1 by -0.004 , all under one percent absolute. Three caveats belong with this. A5 was generated by the template composer that Section 5.2 withdraws, not by the model-generated pipeline, so it measures retrieval’s effect on a generator we no longer report. A5 was also scored on automatic metrics only while A6 was scored by humans, so the pair is not measured on one instrument and the contrast is directional; a human-rated no-retrieval condition is the missing cell. And we withdrew the companion claim that retrieval without the IR scores below a no-context baseline, for the reason given in Section 5.2 (-0.07 , $p = 0.90$). A5 therefore reads simply as retrieval making no measurable difference here, with no accompanying story about it doing harm.

A6, remove the IssueSpec and keep retrieval. This ablation is superseded. Under the confounded design it showed quality falling from 4.59 to 2.24; under the corrected same-model design (Section 5.2) removing the IssueSpec changes quality by -0.04 , which is not distinguishable from zero. A5 and A6 together now say something different from what we first concluded: neither retrieval nor the typed intermediate produces a measurable change in the reply a user receives, and the Stage 3 to Stage 4 coupling we claimed is not supported.

A7, single-stream RLHF feedback. Not run. It was designed to test

985 whether quality and compliance feedback can share one channel, which only becomes a meaningful question once the constraint binds and the two signals compete. Section 5.3 reaches the first half of that condition and not the second, so A7 stays deferred rather than reported.

6. Discussion

990 6.1. What Carries the Result, and What Does Not

Load-bearing. Two components carry the measured gains. Type-routed templating is the principal driver of Stage 3 output: removing taxonomy grounding (A3) drops strict template-fill from 0.96 to 0.69 and collapses the structural per-field metrics to zero. The verified-anchor procedure is the 995 second, and Section 5.1 carries its evidence and its caveats. By contrast the typed IR is *not* a demonstrated driver of Stage 4 quality: under a same-model comparison removing it moves quality by 0.04 points ($p = 0.54$, Section 5.2), removing retrieval instead (A5) moves every automatic metric by less than one point absolute, and the retrieval-versus-no-context claim we made twice 1000 does not survive either (Section 5.2). Nothing about retrieval survives from Stage 4 beyond the null.

Not load-bearing, and reported as such. The knowledge graph does not win on intrinsic cluster geometry. Under count matching (A1b), flat agglomerative clustering beats the KG on all three intrinsic metrics, and sub-cluster 1005 purity is lower at the KG’s finer granularity (A2). The KG’s defensible contribution is centrality-based prioritization, not tighter clusters; we make no cluster-quality claim for it. Even that goes less far than it appears: PageRank runs on the bipartite review-aspect graph, and nothing in our results shows it needs the hierarchical sub-clustering layer. A flat clustering with 1010 the same aspect graph would plausibly support the same ranking, and we did not test that configuration. Stage 2 as built is therefore justified in its prioritization role and unjustified in its sub-clustering role. Stage 5 sits in between: the compliance constraint does bind once the policy is capable of violating it and the dual update engages rather than collapsing, but no 1015 quality-versus-compliance trade-off is demonstrated at the scale we can train at (Section 5.3). It is released as a testbed with a partial result. SPECCOV falls in the same group for a sharper reason: tested against human faithfulness judgements it shows no correlation at all (Section 5.1), so we withdraw it as a measure rather than defend it.

1020 *What this means for the framework claim.* The contribution that survives
ablation is the typed IR and the type-routed templating that produces it,
together with the verified-anchor procedure that makes the IR producible
from noisy labels at corpus scale. Stages 2 and 5 are supporting infrastructure
with honestly mixed evidence. A reader who wants the smallest system that
1025 reproduces what survives needs Stage 1 and Stage 3. Stage 4 is not part of
that, and we state the consequence rather than soften it: on our evidence
Stage 4 contributes nothing we can measure, and a reader building on this
work should treat the reply-generation layer as unproven rather than as a
component with a small effect.

1030 6.2. Implications

Three claims travel beyond app reviews. *First*, LLM annotation noise
on software-engineering text is not uniform, but the shape is not the one we
first reported. We had described the 25% mislabelling rate as concentrated
on minority classes. Recomputing per stratum on the verified anchor shows
1035 the opposite: 1,302 of the 1,307 disagreements fall in the **praise** stratum,
the largest class, at 25.8%, against 0.5% on **performance**. What is true, and
narrower, is that the labeller never emits two of its seven classes at all across
the corpus, 8 **compatibility** and 184 **performance** labels in 215,583 reviews.
Systematic absence of a class, rather than elevated error on rare classes, is
1040 what a small verified anchor is able to repair. *Second*, a caution rather than
a claim. We expected a typed structural intermediate to do the load-bearing
work in spec-aware response generation, and reported it as such; Section 5.2
withdraws that. The only downstream movement we can still point to is
+0.12 on a rule-based scorer from agentic critique-revise iteration, on ten
1045 reviews, which is too small a sample to build on. What generalises is the
negative lesson: a structured intermediate can be demonstrably better as an
artifact and still leave the generated text indistinguishable, and pipelines of
this shape should measure the two separately. *Third*, the count-controlled
cluster-validation ablation (A1b) is a self-contained methodological contribu-
1050 tion: whenever a new clustering design changes cluster count and structure
simultaneously, intrinsic geometry metrics conflate the two effects. The min-
imum reporting standard our analysis suggests is to (a) compare against a
count-matched baseline on the same intrinsic metrics, and (b) pair that with
a domain-utility metric (PageRank coverage in our case).

1055 *6.3. Threats to Validity*

Construct. Template-fill measures coverage rather than content correctness, and Section 5.1 shows that SPECCOV does not fill that gap: it is a lexical-grounding proxy that does not track human faithfulness judgement [47, 48, 49, 50]. The paper therefore has no validated content-
1060 correctness measure for Stage 3, and the five-dimension rubric carries that burden alone. NLI- and QA-based faithfulness measurement [55, 56, 57] is the clear next step. *Internal.* The count-versus-structure confound is exposed and resolved by the A1b ablation, which reframes the knowledge graph’s contribution as centrality-based prioritization rather than intrinsic
1065 geometry. A second internal threat we did not remove is that the 100 clusters carrying the Stage 3 and Stage 4 results were produced by the KG pipeline, the same layer A1b and A2 find wanting. Since flat clustering scores higher on purity, the headline gains may be measured on a slightly worse partition than a simpler system would produce, which would understate
1070 rather than inflate them. Re-running Stages 3 and 4 over flat clusters would settle it and is the cheapest open experiment in the paper. The Stage-1 evidence carries two further internal threats stated with the results: the gold standard is seeded with V5’s predicted label and stratified over V5’s predicted classes (Section 4), and every `compatibility` figure rests on 300
1075 augmented training rows, one internal figure from which we withdraw outright (Section 5.1). *External.* RRGEn contains 58 Android apps, named by real package identifier rather than anonymised; V5 transfers zero-shot to the Maalej label space. Stage 2–4 cross-corpus transfer remains to be validated. The headline LLM is Claude Opus, with Llama-3.3-70B and Qwen2.5-`{3B, 1.5B}`
1080 as cross-family bias controls; GPT-4o and Gemini-1.5-Pro replication is queued (Section 6.5). Every KG-hierarchical number describes a 3.9% sample of the corpus (Section 3.1). *Conclusion.* The three-rater agreement on the classification gold ($\kappa = 0.968$ on the Y/N verdict) sits well above what free-labeling studies of app reviews report, which invites the question
1085 of whether the raters were genuinely independent. Three properties of the design account for the gap. The raters verify a supplied label rather than choose one from seven, which collapses most of the decision space; a 20-review calibration round with adjudicated disagreements precedes the main task; and the written protocol fixes eight boundary rules. Residual
1090 disagreement is rater-specific rather than random, one rater accounting for most of the `usability` reassignments, which is the pattern independent annotation produces. Stage 4 sits differently: all three raters scored the

complete 200-row re-run, and the gain does *not* reproduce per rater (+0.56 against -0.29 and -0.15). That disagreement, not the pooled null, is the honest summary of what we know about Stage 4. *Reliability, and which α belongs to what.* Three agreement statistics appear in this paper and they measure different things, so a single pooled reading would be wrong. The classification gold is annotated by three humans and reaches Fleiss $\kappa = 0.968$ on the verdict and 0.979 on the seven-class label. The Stage-4 response set we report is rated by the same three humans on all 200 rows, with quality weighted κ of 0.03 , 0.04 and 0.50 . That is low, and it is the number to quote: the 0.970 to 0.986 range belongs to the discarded round, where one arm was visibly broken and agreement was easy. The weakest figure, $\alpha = 0.285$ (corrected from a mis-implemented 0.451 , Section 4), comes from a deliberately weaker panel of the lead author plus two Qwen2.5-3B prompts [44]. It bounds how far an LLM rater tracks a human on a subjective judgement; it is not the reliability of any headline result, and no claim in this paper rests on it alone. *Statistical.* Experiment 2 reports a null, so the relevant question is power rather than correction for multiplicity. With 100 paired reviews and the observed spread, the 95% bootstrap interval $[-0.10, +0.18]$ excludes effects larger than about a fifth of a Likert point in either direction. We can rule out a large effect, not a small one. In the other direction, the one cross-family quality difference that reached nominal significance does not survive multiplicity correction and is withdrawn (Section 5.1).

6.4. Reproducibility and Extensibility

Every numerical claim in Section 5 is backed by a saved data file, and the full path from raw RRGGen to Stage 4 output takes approximately six hours on a single GPU-equipped workstation. We audited what a reader can check without that run by execution rather than assumption. Re-running the released analysis scripts regenerates roughly sixty stored result files bit-identically, including every statistical test behind RQ1 and RQ2. What those scripts consume is not itself regenerable from the release: the Stage-3 specification sets, the 100-cluster sample, and the Stage-4 RAG sample ship as artifacts with no standalone producer script, since producing them requires the proprietary-API generation pass. The analysis layer is therefore reproducible from the middle outward and the data-generation layer only by re-running the pipeline against a paid API. One artifact does not pin at all: the GitHub-mined human-written comparison set is re-queried live and returns

1130 a different issue mix on each run, so the `human_github` condition rests on a snapshot we can release but not regenerate.

The release bundles the V1–V5 classifier checkpoints and cleanlab correction procedure; the 5,230-record verified anchor (covering 5,229 distinct corpus rows) and the 490-review classification gold; both Stage-4 rating rounds, 1135 the discarded 400-row one and the reported 200-row re-run, with the generated replies and the rating-provenance check that distinguishes them; the count-controlled A1b output; the SPEC_{COV} scorer with the 60-spec three-rater human faithfulness ratings and the validation analysis; the CMDP-RLHF testbed; the 100-cluster benchmark, with the 99-review LLM-assisted 1140 subsample ($\alpha = 0.285$) kept separate from the all-human agreement sets; a pinned-environment Docker image; and a nineteen-segment verifier that recomputes numerical claims from the saved files, with the discarded Stage-4 round retained as its own segment labelled withdrawn. We audited the verifier itself rather than trusting it, and its coverage is narrower than we 1145 previously stated. Three segments recompute a value from the underlying data and assert it; the remainder load a stored summary and print it, which checks that the manuscript agrees with the released file but not that the file is right. Uncovered numbers include the Stage-3 template-fill table, the held-out κ of 0.616, the Qwen-judge purity figures, the Maalej cross-protocol 1150 row, the per-dimension cells of the cross-family rubric, and most secondary statistics in Sections 5.2 and 5.3. We list this rather than repeat the earlier and much smaller claim that only two tables were uncovered.

Because each stage emits a typed intermediate, components can be substituted independently without breaking downstream contracts. Deterministic 1155 components use fixed seeds; the LLM-dependent stages reproduce statistically rather than bit-identically, and we measured how large that variance is. Regenerating Stage 3 specs three times over the same 15 stratified clusters with the local Qwen2.5-3B generator at temperature 0.7, changing only the seed, gives strict template-fill 0.685, 0.700, and 0.634 (mean 0.673, sd 0.029) 1160 and SPEC_{COV} 2.73, 2.67, and 3.00 (mean 2.80, sd 0.144), with no parse failures. Run-to-run noise on template-fill is therefore around three points on a 0–1 scale, and that is the yardstick a reader should hold up against the differences we report. The gap the paper rests on clears it comfortably, 0.96 versus 0.69 for taxonomy grounding. We previously also cited the 0.96-versus-0.48 1165 cross-family span here; that comparison is not matched on sample or issue type (Table 6), so we no longer use it as a magnitude. The withdrawn Stage-4 effect illustrates the opposite case: at +0.04 it sits an order of magnitude

below this noise floor. Differences of a few hundredths between adjacent conditions, such as 4.09 versus 3.96 at the top of the cross-family rubric, do not clear it and should not be read as a stable ordering. Two limits remain: neither the headline Claude generations nor Stage 4 response generation was repeated, because that needs API budget rather than new code. The released script runs both sweeps unchanged.

6.5. Future Work

Five extensions map to the limitations above, each starting from a released resource. **(0)** Replace SPECCOV with an entailment-based faithfulness measure [55, 56, 57], validated against the 60-spec human ratings we release, and widen the cross-family rubric comparison beyond the 14 matched bug-report clusters with a judge outside all four families. **(1)** Ask the right raters, then scale: the reply from a triage pipeline is consumed by a developer, so the next Stage-4 round should recruit raters with software-engineering experience and no stake in the system before growing the sample from 100 clusters to 200–300. **(2)** Run the CMDP loop at a scale where the trade-off could show, with a longer horizon and a policy with a higher base violation rate on an 8B-class model. **(3)** Frontier-family replication (GPT-4o, Gemini-1.5-Pro) on Stages 3–4. **(4)** Cross-corpus transfer of Stages 2–4; V5 already transfers zero-shot to Maalej [1]. **(5)** An agentic planner over Stage 4 that dispatches a different sub-agent profile per issue type, using SWE-Agent [11], RepairAgent [12], or HyperAgent [30] for defect repair; a spec-level dispatch stub already runs across all five issue types, and end-to-end patch evaluation is the next step.

7. Conclusion

We proposed **IssueSpec**, a knowledge-graph-grounded typed intermediate representation aimed at five gaps: classification stops at the label, response generation skips the structured intermediate, agentic software engineering assumes the spec exists, knowledge graphs emit no typed schema, and RLHF has not addressed dev-rel compliance. We close the first three and report negative results against the last two. A verified-anchor procedure makes the IR producible at corpus scale from noisy LLM labels, lifting Cohen’s κ from 0.163 to 0.592, and to 0.616 on a held-out subset whose value is replay-dependent and indistinguishable from the full-set figure; classifier endorsement of the corrections is 88.66%, a consistency check rather than

independent validation. Cross-family replication on four LLMs preserves the rank order descriptively, with no pairwise difference significant, so it is weak evidence against single-model bias rather than a ruling-out. We set out to show that the typed IR is the load-bearing component of spec-aware response generation, and we cannot: an isolation experiment holding the generator fixed moves human-rated quality by +0.04 Likert points ($p = 0.54$, three raters, 100 paired reviews) with the raters disagreeing in direction, and the +2.35 we previously reported came from a comparison whose two arms were written by different template composers. Nor does the negative half of our original hypothesis survive: we had reported that retrieval alone falls below a no-context baseline, and against the arm that actually has no context it scores 2.24 against 2.31 at $p = 0.90$, inside the discarded round in any case. Stage 4 yields one usable result, the null. Two findings travel beyond app reviews: an LLM labeller on software-engineering text can omit whole classes rather than merely err on them, and a structured intermediate can be demonstrably better as an artifact while leaving the text generated from it indistinguishable, a distinction pipeline evaluations of this shape should measure separately. We release the IssueSpec schema, the verified anchor, the SPECCOV scorer together with the human ratings that show it does not measure faithfulness, the count-controlled cluster-validation ablation, the structured-RAG architecture, the CMDP-RLHF trainers, and the benchmark dataset.

CRediT authorship contribution statement

Fabiha Jalal: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. **Sadia Tasnim Dhruba:** Investigation, Data curation, Validation, Writing – review and editing. **Tazrian Zaman Tushti:** Investigation, Data curation, Validation, Writing – review and editing. **Ajwad Abrar:** Methodology, Supervision, Writing – review and editing. **Md Kamrul Hasan:** Supervision, Resources, Writing – review and editing. **Hasan Mahmud:** Supervision, Resources, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code, configuration files, and analysis scripts are available in the project repository. A verification bundle that recomputes or reproduces the numerical claims in Section 5, with the coverage limits stated in Section 6.4, is archived on Zenodo (DOI 10.5281/zenodo.21982774, v2.0), and the Stage-1 V5 classifier is released on Hugging Face. That bundle supersedes an earlier release (DOI 10.5281/zenodo.20320410) which predates the Stage-4 correction described in Section 5.2; both versions remain resolvable, and the earlier one should not be used. The underlying review corpus derives from the RRGGen dataset of Gao et al. That release is access-gated behind a request form, states academic use only, and carries no licence granting redistribution, so we do not redistribute it. We also do not redistribute the Maalej and Nabil or Guzman corpora. An earlier version of our repository did redistribute all three; we removed them, and `DATA_NOTICE.md` records what was removed and how to obtain each corpus from its original authors. Derived artifacts that embed verbatim review text carry the same restriction as the source and are not offered under an open licence; original code is released under MIT.

Ethics, consent, and data protection

Human participants. Three people annotated for this study: the lead author and two collaborators. The annotation protocol, released with the artifact, states that participation was voluntary, that annotators could withdraw at any time, that no personal information was collected beyond a name and email for attribution, and that compensation took the form of authorship credit. Both additional annotators are consequently co-authors of this paper, which we state explicitly because the paper’s discussion of rater independence should be read with it in mind. The work was not submitted to an institutional review board. Under our institution’s practice a study of this kind, involving colleagues rating text with no collection of sensitive personal data, does not require review; we record that as our reading rather than as a formal exemption, since no exemption number was issued.

Review authors. The corpora we work with are user-generated public reviews whose authors did not consent to research use and cannot practically be contacted. We took three positions on this. We do not redistribute any of the three source corpora, for the licensing reasons in the data-availability statement and because the Maalej and Nabil files contain unredacted email

addresses, telephone numbers, and personal names. Derived files we do re-
release contain review text as RRGGen supplies it, lowercased and lemmatised
1275 with digits masked; that masking is imperfect, and we found a small number
of surviving transaction identifiers in the source, which anyone re-releasing
derived text should re-scan for. We report no result at the level of an in-
dividual review author, and no reviewer identifier appears in any released
file.

1280 *Machine-generated replies.* The replies this system produces read as
human-written. Any deployment should label them as machine-assisted,
which we regard as a condition of use rather than a recommendation.

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the lead-author reference set. The authors verified all technical content,
experimental results, and claims in this paper.

1290 Declaration of Generative AI and AI-assisted technologies in the writing process

We disclose all uses of generative AI in this work in line with Elsevier
policy. *Writing and code assistance.* An earlier version of this disclosure
said that Claude assisted only with grammatical refinement, caption polish-
1295 ing, and LaTeX formatting, and that all pipeline code, experimental scripts,
and analysis were written by the human authors. That understated the use,
and the repository we release contradicts it: 47 of the 56 commits carry
a **Co-Authored-By: Claude** trailer, including commits that add pipeline
stages, the constrained-PPO trainer, the analysis scripts, and the correction
1300 passes described in Section 5.2. The accurate statement is that Anthropic’s
Claude was used throughout implementation and analysis as a coding assis-
tant, under author direction and review, and also for manuscript editing. We
prefer to correct this here rather than leave a disclosure that the artifact re-
futes. Every structural, framing, experimental-design, and claim decision was

made by the authors, and every reported number was produced by author-written code run under human supervision and verified against saved data files. *Models inside the system.* Several generative models are themselves pipeline components and are reported as such in the body: Claude (Opus 4.7) is the headline Stage 3 generator (Table 8); Llama-3.3-70B-Instruct and Qwen2.5-{3B, 1.5B}-Instruct are cross-family Stage 3 comparators (Table 6); Qwen2.5-3B-Instruct is the LLM-as-judge for the five-dimension rubric and the Y/P/N cluster-purity audit, both calibrated against the lead author (Tables 7, 11). The V2 corpus labels are LLM-generated and explicitly treated as noisy, which is what motivates the verified-anchor procedure of Stage 1. *Human verification.* No reported numerical result, expert annotation, or ground-truth label was automatically generated by AI without human verification. The 5,230-review verified anchor, the 490-review classification gold, the 99-review three-rater ratings, and the 20 lead-author IssueSpec references were all produced and reviewed by human annotators (Section 4).

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Appendix A. Implementation Settings

Section 3 states what each stage does and why; this appendix collects
 1545 the exact settings. Stage 1 fine-tunes a RoBERTa multi-label head; confident-learning correction against the verified anchor is gated by anchor confidence ≥ 0.70 and LLM probability ≤ 0.20 , with a 3×3 sensitivity grid over $(0.65, 0.70, 0.75) \times (0.15, 0.20, 0.25)$ reported in Section 5.1. The **compatibility** stratum is seeded with 200 template-generated synthetic
 1550 reviews plus 100 keyword-mined RRGGen reviews, checked by eye without a correction log (Section 3.1). Stage 2 embeds with Sentence-BERT, reduces with UMAP, and clusters with HDBSCAN, aspect-first, producing 605 sub-clusters of mean size 15.9 against 194 flat clusters of mean size 471.0; the KG side runs over an 8,404-review sample and the flat baseline over
 1555 91,368 clustered reviews from the full corpus, so the two describe different review sets. PageRank runs on the bipartite review-aspect graph. Stage 3 routes to one of five templates and re-prompts once with dimension-level feedback on any rubric failure. Stage 4 retrieves over a 15,100-document ChromaDB index built from five sources, with the IssueSpec injected per
 1560 query; the agentic-RAG feasibility study uses 10 stratified reviews, two per

issue type, with Qwen2.5-3B as critic and reviser and up to two iterations. Stage 5 trains five policies on distilGPT2 (82M parameters) from 400 SFT samples, 296 KTO pairs, and 100 DPO pairs; the Lagrangian loop runs 30 steps at batch size 4 with KL coefficient 0.05, and the compliance threshold sweep covers $\tau \in \{0.0, 0.5, 0.7, 0.9, 1.0\}$. The binding-constraint re-run uses Qwen2.5-1.5B-Instruct with LoRA adapters over the attention projections, the frozen base as KL reference, $\tau = 1.0$, and 90 steps, with scorers, dual-ascent rule, and batch size unchanged. SPECCOV counts substantive-token overlap with the source cluster, tokens of length ≥ 5 after stop-word filtering, with quoted-string and coverage bonuses.

Appendix B. Annotator Guidelines

The lead author served as primary annotator across all five annotation tasks. The 490-review classification gold was additionally annotated by two further independent human raters on the identical item set, aligned by review id; all three first completed a shared 20-review calibration round, disagreements were adjudicated before the main task opened, and the released gold label is the majority vote. All human annotation in this work is carried out by those same three people. For Stage 4, all three raters scored all 200 rows of the reported re-run independently and blind to condition; the discarded 400-row round had the lead author as primary rater with one additional rater on a 30-pair subset, and is described here only because the paper reports its withdrawal. Separately, two Qwen2.5-3B-Instruct prompts (concise role-based and chain-of-thought) served as the second and third raters in the 99-review Krippendorff α calibration, which is an LLM-assisted panel and not one of the human rounds. For the 5,230-record verified anchor, annotators classified each review into one of seven classes (**bug_report**, **feature_request**, **performance**, **usability**, **compatibility**, **praise**, **other**) with multi-label assignment permitted; reviews below the “fairly confident” threshold were skipped. For the 490-review gold, a stratified sample of 70 reviews per class was drawn and the lead author re-labeled each review blinded to the V2 LLM prediction. For the 5-dimension IssueSpec rubric, each spec was scored 1–5 on *completeness* (are all required template fields filled?), *accuracy* (do field values reflect the source cluster?), *actionability* (would a developer know what to do?), *specificity* (does the spec name concrete entities?), and *clarity*, each dimension scored independently. For the reported Stage 4 evaluation (100 reviews $\times$ 2 conditions), all three raters rated *quality* (1–5),

specificity (1–5), and *helpful* (Y/N) while blinded to the generation method. In both rounds condition labels were replaced with random codes and unsealed only at aggregation, with the assignment persisted as spreadsheets with JSON master keys. For the Y/P/N cluster-purity rubric, annotators read 5 representative reviews per cluster and assigned *Y* (all 5 share the sub-theme), *P* (3 or 4 of 5 share), or *N* (fewer than 3 share); weighted purity is $\text{purity}_w = (|Y| + 0.5|P|) / (|Y| + |P| + |N|)$. All comparative evaluations were blinded throughout the rating window; reviews or specs below the annotator confidence threshold were marked “skip” and excluded.

Table 3: IssueSpec standardized issue schema. Each issue type follows the dominant industry or HCI standard. Each template is drawn from a published standard or reference and reproduces its named fields; we ran no conformance check against the standards documents themselves, so “drawn from” should not be read as “certified against”.

Issue type	Template (citation)	Justification
bug_report	title, severity, steps_to_reproduce, expected, actual, component	The tracker-standard field set in GitHub Issues and JIRA. Chaparro et al. [4] support three of the six (observed behaviour, expected behaviour, steps to reproduce). We previously attributed the whole set to Zimmermann et al. [5]; that was wrong twice over. They propose no template, and in their developer survey severity ranks last at 0% importance and component at 3%, against 83% for steps to reproduce. We keep both fields because trackers require them, not because developers report finding them useful
feature_request	John user-story [40] (As-a / I-want / So-that) + behavior-driven development (BDD) acceptance criteria [41]	Dominant industry format (Agile, Scrum, SAFe); BDD standardizes testable outcomes
performance	Fields are latency, startup_time, memory_consumption, battery_drain, network_usage, mapped onto ISO/IEC 25010 [42] performance-efficiency sub-characteristics: the first two to time behaviour, the last three to resource utilization; capacity is not covered	International standard for software-quality NFR taxonomy. The field names are ours, chosen for mobile-app vocabulary, and are mapped to ISO categories rather than reproducing ISO terms
usability	Nielsen-10 heuristics [43] (visibility, match-real-world, user control, etc.)	Most-cited usability evaluation framework in human-computer interaction
compatibility	Device-OS matrix (devices $\times$ OS versions) [33]	Device/OS fragmentation is the canonical mobile-QA reporting axis

Table 4: Annotator-vs-LLM and inter-annotator deviation. The $\approx 25\%$ deviation is measured on the human-verified anchor; the κ -progression reaches moderate agreement [51]; the V5 endorsement row is a consistency check, not independent validation, since V5 trained on those corrections; LLM-judge calibration Δ bounds judge bias. The $\alpha = 0.285$ row is an LLM-assisted panel, corrected from a mis-implemented 0.451 as described in the text.

Annotator pair	sample	agreement / deviation
Lead-author vs V2 LLM	5,230	75% / 25% mis-labeled
Lead vs V2 / cleanlab / V5	489 gold	κ : 0.163 $\rightarrow$ 0.333 $\rightarrow$ 0.592
same, vs 3-rater majority gold	489 gold	κ : 0.165 $\rightarrow$ 0.334 $\rightarrow$ 0.590
3 human raters, Y/N verdict	490 gold	Fleiss $\kappa =$ 0.968 ; 479/490 unanimous on the verdict, 476/489 on the label
3 human raters, 7-class label	489 gold	Fleiss $\kappa =$ 0.979 ; $\alpha = 0.979$
V2 LLM vs V5; cleanlab vs V5	215,583	71.96% / 86.77% agree
cleanlab vs V5 (consistency, not independent)	40,291	88.66% support
V5 vs gold, held out (replay-dependent, see §5.1)	307	$\kappa =$ 0.616
3 human raters (Stage 4 quality, discarded round)	400 rated rows	weighted κ 0.970–0.986; within-1 100%
3 human raters (Stage 4 quality, reported round)	200 rated rows	weighted κ 0.03, 0.04, 0.50
3 human raters (Stage 4 helpful, discarded round)	400 rated rows	κ 0.678–0.836; $\alpha = 0.781$
3-rater <i>LLM-assisted</i> classification check (lead + 2 Qwen prompts)	99 re-views	Krippendorff $\alpha = 0.285$
Lead vs Qwen-judge (purity, rubric)	50 / 28	$\Delta = -0.035$ / +0.23

Table 5: Cross-family five-dimension rubric, same Qwen2.5-3B judge as Table 7, on matched clusters. Panel A is the four-way comparison on clusters covered by every generator (bug reports only); none of its six pairwise differences is significant at $n = 14$ (Wilcoxon p from 0.18 to 0.75), so read it as descriptive. Panel B widens the Claude-versus-Llama comparison to all five issue types and gives a nominal +0.38 at Wilcoxon $p = 0.015$; we withdraw that as a claim, because it fails Bonferroni or Holm correction over the seven tests in this table ($p = 0.104$) and an exact sign test ($p = 0.078$). Read both panels as descriptive.

generator	n	mean	compl.	accur.	action.	spec.	clarity
<i>Panel A, four-way, 14 matched bug-report clusters</i>							
Claude (Anthropic)	14	4.09	4.00	4.50	3.71	4.36	3.86
Llama-3.3-70B (Meta)	14	3.96	3.86	4.50	4.07	3.86	3.50
Qwen2.5-3B (Alibaba)	14	3.83	3.07	4.50	4.00	3.86	3.71
Qwen2.5-1.5B (Alibaba)	13	3.77	3.00	4.15	4.46	3.62	3.62
<i>Panel B, Claude vs Llama, 28 clusters, all five issue types</i>							
Claude (Anthropic)	28	4.01	3.86	4.54	3.75	4.29	3.64
Llama-3.3-70B (Meta)	28	3.64	3.39	4.46	3.14	3.82	3.36

Table 6: Multi-LLM evidence summary across three model families (Anthropic, Meta, Alibaba). Cross-family rank preservation in Stage 3 is the structural-bias control; LLM-as-judge calibration Δ vs the lead author bounds judge bias. *Two rows carry caveats a reader needs.* The Stage-4 row previously reported 0.61 versus 0.54 for “Claude” against Qwen-3B. We withdraw it: the file labelled Claude is `responses_rrgen_baseline.json`, which our own release notes record as the output of a deterministic template composer rather than of any language model, so the row compared a template against a generation. This is the same confound we withdrew in Experiment 2 and we had not noticed it here. The Stage-3 row is a real cross-family comparison but is not matched: Claude is scored on 100 clusters and Llama on 99, both spanning five issue types, whereas both Qwen conditions are 15 clusters of `bug_report` only. The 0.96-versus-0.48 span therefore compares a five-type mean against a single-type mean and should be read as an ordering, not as a magnitude.

Use		Model (family)	Result
Stage 3 generate		Claude (Anthropic) / Llama-3.3-70B (Meta) / Qwen-3B (Alibaba) / Qwen-1.5B (Alibaba)	0.96 / 0.80 / 0.74 / 0.48
Stage 4 generate		<i>withdrawn</i> (see <i>withdrawn</i> note)	
5-dim judge	rubric	Qwen-3B, all specs	100 3.94/5 (28-spec subsample: 3.89)
Cross-family rubric		Claude / Llama / Qwen-3B / Qwen-1.5B	4.09 / 3.96 / 3.83 / 3.77 (Table 5)
Y/P/N judge	purity	Qwen-3B vs lead-author	0.625 vs 0.66; $\Delta = -0.035$

Table 7: Experiment 1 five-dimension rubric (Qwen-judge, 1–5 scale) on the *complete* spec set: all 100 taxonomy specs and all 20 lead-author reference specs, scored in one run so the two columns are directly comparable. The middle column repeats the earlier stratified 28-spec subsample under the same judge and rubric; the gap between it and the full set is +0.05, which is the subsample error. The LLM-as-judge protocol follows the calibration-against-reference convention of [44, 45, 46].

dimension	taxonomy, $n = 100$	taxonomy, $n = 28$	lead-author ref, $n = 20$
completeness	3.72	3.79	3.20
accuracy	4.43	4.39	4.10
actionability	3.86	3.71	2.85
specificity	4.15	4.07	4.05
clarity	3.52	3.50	3.05
overall mean	3.94	3.89	3.45
sd across specs	0.59	—	0.41

Table 8: Experiment 1 substantive template-fill (strict criteria). The 0.96 vs 0.53 LLM-vs-GitHub gap is coverage, not content quality; the strict-vs-very-strict sweep preserves ordering. The exact-zero cells in **bugs:repro** and **features:story** columns for (b), (c), (d), and (e) are structural: free-form LLM, raw concatenation, lead-author, and human-GitHub specs do not emit the template-specific reproduction-step list or As-a/I-want/So-that triple at all when not prompted for them, so the strict-fill rate is zero by construction.

condition	strict	bugs:repro	features:story
(a) LLM + taxonomy	0.96	0.73	0.97
(b) LLM free-form	0.69	0.00	0.00
(c) raw concat	0.34	0.00	0.00
(d) lead-author ref ($n = 20$)	0.69	0.00	0.00
(e) human GitHub (3 repos)	0.53	0.13	0.00

Table 9: Per-class F1 against the 490-review expert gold. V2 LLM is blind to the two minority classes (compatibility, performance). The verified-anchor procedure with targeted augmentation recovers them; as the text notes, the `compatibility` row rests on 300 augmented training rows rather than on natural corpus evidence.

class	V2 LLM	cleanlab-V2	V5
<code>bug_report</code>	0.42	0.45	0.58
<code>feature_request</code>	0.32	0.37	0.57
<code>performance</code>	0.00	0.47	0.77
<code>usability</code>	0.11	0.27	0.55
<code>compatibility</code>	0.00	0.00	0.83
<code>praise</code>	0.38	0.62	0.68
<code>other</code>	0.30	0.47	0.60
macro F1	0.22	0.38	0.65

Table 10: Stage-4 re-run, same model with and without the IssueSpec, 100 paired reviews, three raters, all 200 rows scored by each. All figures are unadjusted, so the gain column equals the difference of the two condition columns; slot-stratified values are in the released JSON and differ by at most 0.03. The pooled effect is null, and the disagreement between the lead author and the two independent raters is larger than the effect being measured.

rater	no_spec	full	gain	95% CI	Wilcoxon p
lead author	4.04	4.60	+0.56	[+0.44, +0.68]	2×10^{-11}
independent rater 1	3.87	3.58	-0.29	[-0.56, -0.01]	0.06
independent rater 2	3.57	3.42	-0.15	[-0.33, +0.03]	0.09
pooled	3.83	3.87	+0.04	[-0.10, +0.18]	0.54

Table 11: Stage 2 designs head-to-head, comparing Pure HDBSCAN (without UMAP), original flat UMAP+HDBSCAN, and KG-hierarchical. Intrinsic metrics (Davies-Bouldin [58], Calinski-Harabasz [59], silhouette [60]) follow standard convention [61, 62, 63]. The Y/P/N weighted-purity rubric is a discrete-coding instantiation of the extrinsic cluster-purity metric [64, 61]. The two flat intrinsic cells are withdrawn for the alignment fault described in the text; the count-controlled flat baseline (agglomerative-605) is reported in the A1b ablation (§5.5).

metric	Pure HDB	Flat U+H	KG hier.
n_clusters	3 (deg.)	194	605
mean cluster size	~16,933	471.0	15.9
Davies-Bouldin ($\downarrow$)	n/a	<i>withdrawn</i>	2.24
Calinski-Harabasz ($\uparrow$)	n/a	<i>withdrawn</i>	1.85
Y/P/N purity (lead-author)	n/a	0.66	not run
Y/P/N purity (Qwen-judge)	n/a	0.625	0.390